

Theory and Methodology

Multiple tests for validation of system dynamics type of simulation models *

Yaman BARLAS

Systems Analysis Department Miami University Oxford, OH 45056, USA

Abstract: Validation of model behavior is an important part of simulation validation in general, System Dynamics model validation in particular. System Dynamics methodology has often been criticized for its lack of quantitative/formal behavior evaluation tools. Indeed, System Dynamics models have certain characteristics that render standard statistical tests inappropriate. But we argue that such inappropriateness should not lead to adopting entirely qualitative procedures. The challenge is to design novel quantitative tests that are appropriate for System Dynamics behavior evaluation. Such tests should focus on major time *patterns* rather than individual data points.

We suggest a set of procedures that focus on major pattern components of a given composite behavior pattern. We then investigate the appropriateness and properties of those procedures through a series of simulation experiments. The experiments show that the suggested procedures are easy to apply and interpret and suitable for System Dynamics type of behavior evaluation.

Keywords: Systems, simulation, testing, system dynamics, model validation

1. Introduction

Validation of simulation models in general, and System Dynamics (SD) models in particular consists of two types of validity tests:

(1) *Structural* validity tests, the function of which is to check whether the structure of the model is an adequate representation of the real structure.

(2) *Behavior* validity tests, the function of which is to check if the model is capable of producing an acceptable output behavior.

The way these two types of tests are related to overall SD model validity is shown in Figure 1. As

seen in the Figure, direct structural validation is typically achieved in two parts: (i) By comparing the model equations with the real system relationships ('empirical' structural validation) and (ii) by comparing the model equations with the available theory ('theoretical' structural validation). Similarly, behavior validation consists of two parts: (i) Tests for *behavior pattern prediction*: These tests try to determine whether the behavior patterns generated by the model are close enough to the major patterns exhibited by the real system. (ii) Behavior tests that are *structurally oriented*: By examining the model behavior under different conditions, these tests try to determine whether there is a major error in the *structure* of the model. (Such as testing the plausibility of the model behavior under certain extreme conditions.)

The above four types of validity testing must be applied in a certain logical order. First, the model

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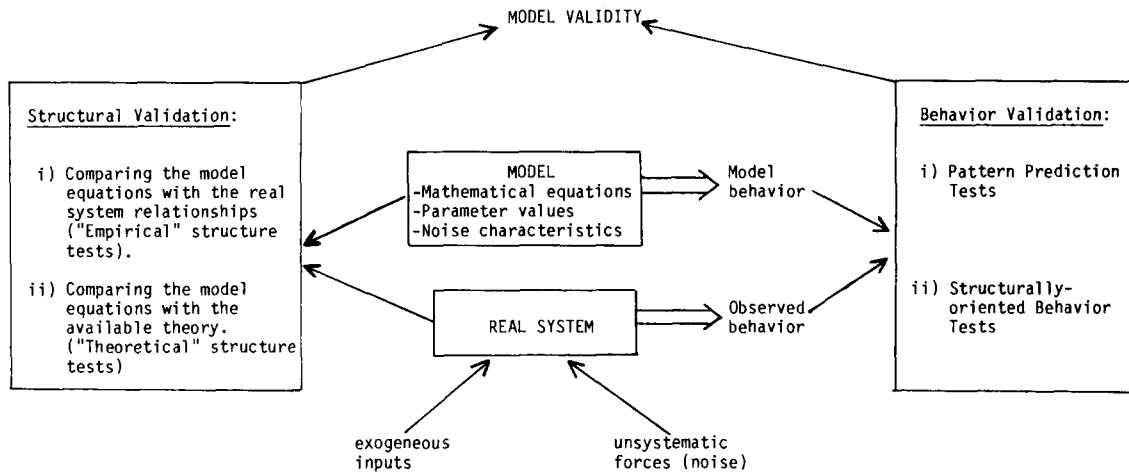


Figure 1. Major aspects of system dynamics model validation

must pass the *structural* tests, then the *structurally-oriented* behavior tests, and finally the *behavior pattern prediction* tests. The reasoning behind this ordering is that unless we are confident that a SD model is structurally sound, there is no point in testing it for pattern prediction ability. This is a corollary of the fact that a model with obvious structural errors can, with enough 'parameter tuning', generate extremely accurate behavior predictions. Thus, pattern prediction tests alone can *not* separate this type of 'spurious' behavior accuracy from 'true' behavior accuracy. On the other hand, if we *know* that the model is structurally sound, then a weak pattern prediction ability indicates that some parameter values or exogeneous inputs are misrepresented in the model, whereas a strong pattern prediction adds to our confidence in the usefulness of the model.

In this article, we focus specifically on tests of *pattern prediction*. This problem is important for two reasons: First, standard statistical tests are not applicable to the SD-type of *pattern* predictions; and second, one of the most frequent criticisms of SD methodology is that it lacks formal/quantitative predictive ability tests. There is a need for formal, quantitative yet practical tests.

2. Pattern prediction testing

Pattern Prediction testing basically consists of comparing the model-generated behavior to the observed behavior. Such a comparison is difficult and problematic for SD type of models for several

reasons: Noise terms of SD models are not necessarily independent and/or normally distributed; behavior patterns generated by typical SD models are highly autocorrelated, and they are in general non-stationary in the mean and sometimes non-stationary in the variance. Such characteristics of SD models violate the three fundamental assumptions of standard statistical tests: normality, independence and stationarity. Therefore, standard hypothesis tests (such as the *t*-test, F-test, χ^2 -test) are not directly applicable to SD behavior validation. Faced with these problems, one approach would be to 'remove' the undesirable characteristics (such as autocorrelation, non-stationarity) from the data, and then apply the standard tests to the transformed data. But this approach is inappropriate for SD behavior validation, because these statistically undesirable characteristics are the very *time pattern* components (cycles, trends, etc.) that the SD model is expected to reproduce. Instead of removing them, the analyst is responsible for demonstrating that the model is capable of generating autocorrelations and non-stationarities (such as cycles, growth trends, declines) similar to the ones observed in the real behavior. It is well demonstrated that, by their nature, SD models do *not* predict specific *individual* values of output variables ('point prediction'), but they predict major *time patterns* of concern ('pattern prediction'). (See for instance Forrester [8,9], Forrester and Senge [11], Sterman [18], Wright [20].)

As a result of the inappropriateness of the standard statistical tests, System Dynamicists have adopted relatively qualitative procedures for be-

havior pattern validation. SD has been frequently criticized for its lack of quantitative/formal behavior evaluation tools. (See, for example, Ansoff and Slevin [3].) SD practitioners have responded to such criticisms by pointing out that model validation is inherently *relative* and *judgemental*, hence largely informal and qualitative by its nature (see Forrester [9]). But Sterman [18] notes that the reluctance to use formal quantitative tools creates an “impression of sloppiness and unprofessionalism”. Making full use of formal quantitative tools of behavior evaluation is *not* inconsistent with a relativist/holistic philosophy of model validation. Thus, the challenge for System Dynamicists is to design novel quantitative tests that are appropriate for behavior *pattern* evaluation. There has been some limited effort in this area (Wright [20], Forrester and Senge [11], Sterman [18]).

In this paper, we suggest a set of *pattern-oriented* measures appropriate for SD behavior validation. We identify five major components present in complex behavior patterns: *Trends, periods of oscillations, phases of oscillations, average values and amplitudes* (variations). We show that a proper behavior evaluation method should measure these pattern components individually, one at a time.

3. The suggested multiple tests

In designing a test for behavior validation, two major criteria must be taken into account:

(1) *Appropriateness*: As stated earlier, for SD type of behavior validation, the test must be pattern oriented, rather than point oriented.

(2) *Simplicity*: The test must be relatively easy to implement and interpret. It is unlikely that an extremely complicated procedure will be widely used by practitioners.

The challenge is to design relatively *simple* tests that are *appropriate* for SD-type of behavior pattern evaluation. Although there is a variety of behavior tests available in statistical literature, most of them will fail with respect to one of the two criteria.

To illustrate the point, consider using Box–Jenkins type of (ARIMA) models as behavior pattern tests. Since ARIMA models are capable of handling nonstationarities, autocorrelations and cyclic behavior, they certainly qualify as ‘ap-

propriate’ for SD behavior pattern evaluation. To implement such a test, one would have to fit one ARIMA model to the model behavior and a second ARIMA model to the real system behavior, and then compare the two models. This means that for each ARIMA model, one would go through the standard ARIMA model ‘identification’, ‘estimation’ and ‘validation’ steps. The problem is, none of these steps is ‘simple’. In dealing with non-stationary and cyclic data, model identification and validation steps could turn out to be quite complicated and even problematic. But the analyst who tries to resolve the validity of an SD model would not want to face this equally challenging ARIMA modeling/validation task! Thus, using ARIMA models as behavior validation tools would score high with respect to the ‘appropriateness’ criterion, but low with respect to the ‘simplicity’ criterion.

Now, to give an example at the other extreme, consider the coefficient of determination R^2 as a SD behavior pattern prediction test. R^2 is perhaps the simplest and the most widely used validation test available, but it would score very low with respect to the ‘appropriateness’ criterion. It is extremely inappropriate for SD-type of *pattern* evaluation, because it measures the discrepancies between the model behavior and the real behavior on a point-by-point basis.

The tools suggested in this article are chosen and designed by keeping the two criteria in mind. These tools are not only appropriate for SD-type of *pattern* evaluation, but they are also reasonably *simple*. The following list, in the given order, describes the suggested multi-step behavior pattern validation procedure:

3.1. Trend comparison and removal

Since most statistical procedures require stationarity in the means, the first step is to estimate, compare and then remove the trend components (if any). As a simple procedure, we suggest fitting regression lines to the model-generated and observed trends. A linear trend can be estimated by $\hat{Y} = b_0 + b_1t$, a quadratic trend by $\hat{Y} = b_0 + b_1t + b_2t^2$, or an exponential one by $\hat{Y} = b_0 e^{bt}$. The analyst should choose the proper form of equation depending on what type of trend the plotted behavior indicates. Once the proper equation is fit to the observed and model-generated behavior pat-

terns, the estimated coefficients b_i must be compared to see if there is a significant trend difference between the two time patterns. If there is, then the necessary model modifications are called for. After deciding that no significant difference remains, the trend components must be removed by applying the transformation $Z_i = Y_i - \hat{Y}_i$ to both the observed and model-generated patterns.

3.2. Autocorrelation function test for period comparison

Autocorrelation functions are used extensively in estimation of time series models (Box and Jenkins [6]); but they have not been used in validation of simulation models. Spectral analysis, which is equivalent to autocorrelation analysis in the frequency domain, has been proposed as a validation tool by several authors (see for instance Naylor, et al. (13)). Spectral analysis is very useful in detecting the different high frequency cycles that exist in 'engineering' type of time histories. SD type of histories on the other hand do not typically have many intrinsic high frequency cycles. Autocorrelation analysis, which is simpler, cheaper and easier to interpret, seems to be more appropriate for SD type of behavior pattern validation. We design a test based on the sample autocorrelation functions to compare the autocorrelations of the observed and model-generated behavior patterns. Specifically, the test can be used to detect significant errors in the *periods* of the two behavior patterns. The 'sample autocovariance function' of a time series X_i is given by

$$\text{Cov}(k) = \frac{1}{N} \sum_{i=1}^{N-k} (x_i - \bar{x})(x_{i+k} - \bar{x})$$

for lag $k = 0, 1, 2, \dots, N$, where N is the number of data points. The 'sample autocorrelation function' is obtained by dividing $\text{Cov}(k)$ by $\text{Cov}(0)$ (sample variance of X_i):

$$r(k) = \frac{\text{Cov}(k)}{\text{Cov}(0)} = \frac{\text{Cov}(k)}{\text{Var}(X_i)}$$

for lag $k = 0, 1, 2, \dots, N$.

This function gives a measure of how the successive data points are interrelated. We plot $r(k)$ vs. k only for positive lags 1, 2, 3... because $r(k) = r(-k)$. The function is always 1 at lag 0, and lies between +1 and -1 at all other lags

(since it is 'normalized'). As k becomes large and larger, $r(k)$ approaches 0. For random (or 'almost random') series, $r(k)$ immediately drops to 0 for $k \geq 1$. For moderately correlated series, $r(k)$ approaches 0 in a negative exponential fashion. If the series involves trend (non-stationary in the mean), $r(k)$ dies down very slowly; if the series is cyclic, so is $r(k)$. (See Box and Jenkins [6] for a thorough discussion of these properties.)

To be able to carry out a formal test on the equality of the autocorrelation functions of the observed and model-generated time patterns, we need the sampling distribution of $r(k)$. The latter is unknown except for some extremely restricted cases (See T.W. Anderson [2].) As an approximation, we can compute the standard errors (Se) of $r(k)$ and construct a 2Se confidence band around the differences of the two autocorrelation functions. A number of variance estimates are available for $\text{Var}(r(k))$, the most popular being Bartlett's formula [5]. After some empirical analysis, we have found that the only finite version that is general enough and practical for our purpose is the one provided by O.D. Anderson [1]:

$$\begin{aligned} \text{Var}(r(k)) &= \frac{1}{N(N+2)} \sum_{i=1}^{N-1} (N-i)(r(k-i) + r(k+i) \\ &\quad - 2r(k)r(i))^2. \end{aligned}$$

Using this estimator, we can devise an approximate hypothesis test: If $r_S(k)$ belongs to the simulation output and $r_A(k)$ to the actual (observed) one, then the null hypothesis is

$$\begin{aligned} H_0: \quad r_S(1) - r_A(1) &= 0, \quad r_S(2) - r_A(2) = 0, \dots, \\ r_S(M) - r_A(M) &= 0, \end{aligned}$$

and the alternative hypothesis is

$$H_1: \quad r_S(k) - r_A(k) \neq 0 \quad \text{for at least one } k.$$

Now consider the difference $d_1 = r_S(1) - r_A(1)$. The standard error of d_1 is

$$\text{Se}(d_1) = \sqrt{\text{Var}(r_S(1)) + \text{Var}(r_A(1))}.$$

Since $d_1 = 0$ under H_0 , we construct the interval $\{-2\text{Se}(d_1), 2\text{Se}(d_1)\}$ and reject H_0 if d_1 falls outside the interval. As we repeat this procedure for every $k = 1, 2, \dots, M$, we obtain a $2\text{Se}(d_k)$ confidence band, and reject H_0 if any d_k falls

outside the band. Under normality assumptions, this constitutes a hypothesis test with $\alpha = 0.05$ approximately. A major source of approximation is the fact that H_0 is a multiple test and since successive $r(k)$'s are typically dependent random variables, computation of the 'true' significance level requires estimation of the k -dimensional joint distribution of $r(k)$, which represents a very difficult problem in practice. (See T.W. Anderson [2] for a thorough discussion of the issue.) We believe that many other errors inherent in behavior validation are of much larger magnitude than this imprecision of α . For these reasons, trying to improve the precision of α seems neither reasonable nor necessary. However, the analyst should keep in mind that, in general, if (s)he chooses to fix the significance level for the *overall* test at a certain level (say 0.05), then the levels for the *individual* tests should be set at a smaller value (say $\alpha = 0.01$).

3.3. Cross-correlation function test for phase lag detection

The cross-correlation function between the two time series shows how the two series are correlated at different time lags. If the two series are perfectly in phase, this function is maximum at lag 0; if the two series are out-of-phase the function reaches its maximum at some non-zero lag. The cross-correlation function between the simulated (S) and the actual (A) time patterns is given by:

$$C_{SA}(k) = \frac{(1/N) \sum_{i=-k}^N (S_i - \bar{S})(A_{i-k} - \bar{A})}{s_S s_A}$$

for $k = 0, 1, 2, \dots$

and

$$C_{SA}(k) = \frac{(1/N) \sum_{i=-k}^N (A_i - \bar{A})(S_{i+k} - \bar{S})}{s_S s_A}$$

for $k = 0, -1, -2, \dots$,

where

$$\bar{S} = \frac{1}{N} \sum_{i=1}^N S_i, \quad \bar{A} = \frac{1}{N} \sum_{i=1}^N A_i,$$

$$s_S = \sqrt{\frac{1}{N} \sum (S_i - \bar{S})^2},$$

$$s_A = \sqrt{\frac{1}{N} \sum (A_i - \bar{A})^2}.$$

The cross-correlation function is not necessarily symmetrical. But note that $C_{SA}(k) = C_{AS}(-k)$. Thus, the function is fully known when $C_{SA}(k)$ and $C_{AS}(k)$ are known, or when only one of them is known for both positive and negative lags. Note also that $C_{SA}(0) = C_{AS}(0)$ which is nothing but the simple correlation coefficient between the two series.

In the following sections, we will show that by comparing the cross-correlation functions of the simulated and observed behavior patterns, we can decide whether—and by how much—the two behavior patterns are out of phase.

3.4. Comparing the means

To compare the means of the simulated and observed behavior patterns, we compute the sample means S and A as defined above. These two sample means could be compared using a standard t -test, but the problem is that both S_i and A_i are in general highly autocorrelated data sets. As a practical alternative approach we can define the 'Percent error in the means' E1 as

$$E1 = \frac{|\bar{S} - \bar{A}|}{|\bar{A}|}$$

and examine this quantity to see if there is a large discrepancy between the means. Experimental results will give us an idea about 'how large is large'.

3.5. Comparing the amplitude variations

To compare the variations in the simulated output, we make use of the standard deviations and define the 'Percent Error in the Variations' E2 as

$$E2 = \frac{|s_S - s_A|}{s_A}.$$

Some SD behavior patterns exhibit a very obvious dominant cycle. In those cases, in addition to the sample estimates of variations (such as E2), the amplitudes of the dominant cycles should be graphically measured and compared.

3.6. Summary measure of overall behavior discrepancy

We have stated earlier in this article that single summary measures are not very informative in

evaluating the validity of model behavior. On the other hand, single measures (such as R^2) are preferred by most practitioners and readers as simple vehicles of *communicating* the results. With this in mind, but with great caution, we suggest a ‘Discrepancy Coefficient’ to be used strictly as the *last step* of behavior evaluation, only after the model has passed all the previous tests of validity. In such a usage, the ‘Discrepancy Coefficient’ will serve *not* as a test procedure, but as a ‘summarizing and reporting’ tool.

The Discrepancy Coefficient is a version of Henry Theil’s ‘inequality coefficient’ [19]:

$$U_0 = \frac{\sqrt{\sum (S_i - A_i)^2}}{\sqrt{\sum S_i^2} + \sqrt{\sum A_i^2}}.$$

The disadvantage of this coefficient, as noted by Theil himself, is that its value is not uniquely determined by the Sum of Squared Errors (SSE). Of two different sets of predictions with exactly the same SSE, the one with larger S_i would yield smaller U_0 . It is even possible to have two sets of predictions such that $SSE_1 > SSE_2$, but $U_1 < U_2$. As a way of solving this problem, Griner et al. [12] suggest centering the data by subtracting from S_i and A_i their respective means:

$$\begin{aligned} U &= \frac{\sqrt{\sum (S_i - \bar{S} - A_i + \bar{A})^2}}{\sqrt{\sum (A_i - \bar{A})^2} + \sqrt{\sum (S_i - \bar{S})^2}} \\ &= \frac{\sqrt{\sum (E_i - \bar{E})^2}}{\sqrt{\sum (A_i - \bar{A})^2} + \sqrt{\sum (S_i - \bar{S})^2}} \\ &= \frac{s_E}{s_A + s_S}. \end{aligned}$$

We use this ‘Discrepancy Coefficient’ U , which is insensitive to additive constants and preserves its nice property of being between 0 (‘perfect predictions’) and 1 (‘worst predictions’).

Each step of the above suggested multi-step procedure focuses on a different pattern component. But the outcomes of the different steps are *not* entirely independent of each other. These interdependencies will be demonstrated in the experimental results section. For example, if the

model behavior has a significant error in the trend component, then the measures of the ‘error in the means’ and the ‘error in the variances’ lose their meaning. Similarly, if the model behavior has a significant error in the period component (i.e. the ‘autocorrelation test’ not passed), then the cross-correlation function/phase lag test is *not* interpretable. Due to such interdependencies, the six-step procedure must be implemented in the given *logical sequence* in order to be most informative.

4. Research procedure

The next step of this research is to test the performance of the multi-step behavior validation procedure described above. For this purpose, we would like to run controlled experiments to see what types of behavior errors, under what conditions, are detectable by which tests, and what numerical values to expect from various measures. It is not possible to answer such questions by using models of a real system, because it is practically impossible to run controlled experiments with SD-type of real Systems (due to complexity, time frame involved, dimensions, etc.). Therefore, we take a ‘synthetic’ experimental approach similar to Senge’s [17]. We first build a SD model (with structure ‘R’) and call it the ‘synthetic real system’. We then build a model (with structure ‘M’) of the synthetic real system. The synthetic system generates the ‘real’ behavior patterns P_R and the model generates its behavior patterns P_M . This makes it possible to run controlled simulation experiments and test the performance of the multi-step validation procedure, because we have perfect control over the ‘real’ system.

In this article, as was explained in the Introduction, we focus on testing the *pattern prediction ability* (rather than structural validity) of SD models. For this purpose, we assume that the model has *already passed* the structural validation phase. In the experiments presented below, we test the performance of the suggested multi-step procedure in face of non-structural errors such as system noise, input errors, parameter errors, and observation errors. (For similar experiments with structural errors and ‘structurally-oriented’ behavior tests see Barlas [4].)

5. The synthetic system

As the 'synthetic reality', we specify a set of relationships believed to be a realistic description of certain *epidemic dynamics*. The flow diagram of this synthetic system is given in Figure 2. It can be verbally described as follows: When a 'susceptible' person contacts a 'contagious' person, (s)he is infected with some probability. After a person is infected, (s)he becomes 'contagious' for some period of time during which (s)he can transmit disease. At the end of this period the person starts showing symptoms and is recognized as contagious, and (s)he is isolated (or people minimize their contacts) until (s)he recovers or dies. After recovery, the person becomes immune for a period of time, at the end of which (s)he completes the cycle by becoming susceptible again. The second source of replenishment of susceptible population is the 'birth rate'. Every sub-population contributes to the birth rate except the 'Contagious Population Recognizable', which is assumed to exercise some sort of contraception. When infants are born, they are immune for a certain period of time before they become susceptible.

The equations that correspond to the flow diagram of Figure 2 are given in Appendix 1.

Accordingly, the larger the contagious population, the greater the number of infective contacts, the larger the infection rate (a *positive* feedback loop). But the larger the infection rate, the smaller becomes the susceptible population which in turn means a smaller infection rate (a *negative* feedback loop). Thus, the contagious population first goes up, and then declines. As the susceptible population is replenished through births, this growth-then-decline pattern repeats itself, yielding the epidemic cycles (Figure 3). These epidemic oscillations (seen in Figure 3) are 'noisy' rather than deterministic, because some system relationships have random components.

In the following paragraph, we describe only those equations that have *random* components, because the simulation experiments are obtained by changing those equations. (See Appendix 1 for the complete list of equations.) The first place where noise is introduced is the formulation of the 'infection rate' (IR):

$$IR = IPC * CR \text{ (people/month)},$$

IPC: *Infections per contact*,

CR: *Contact rate*.

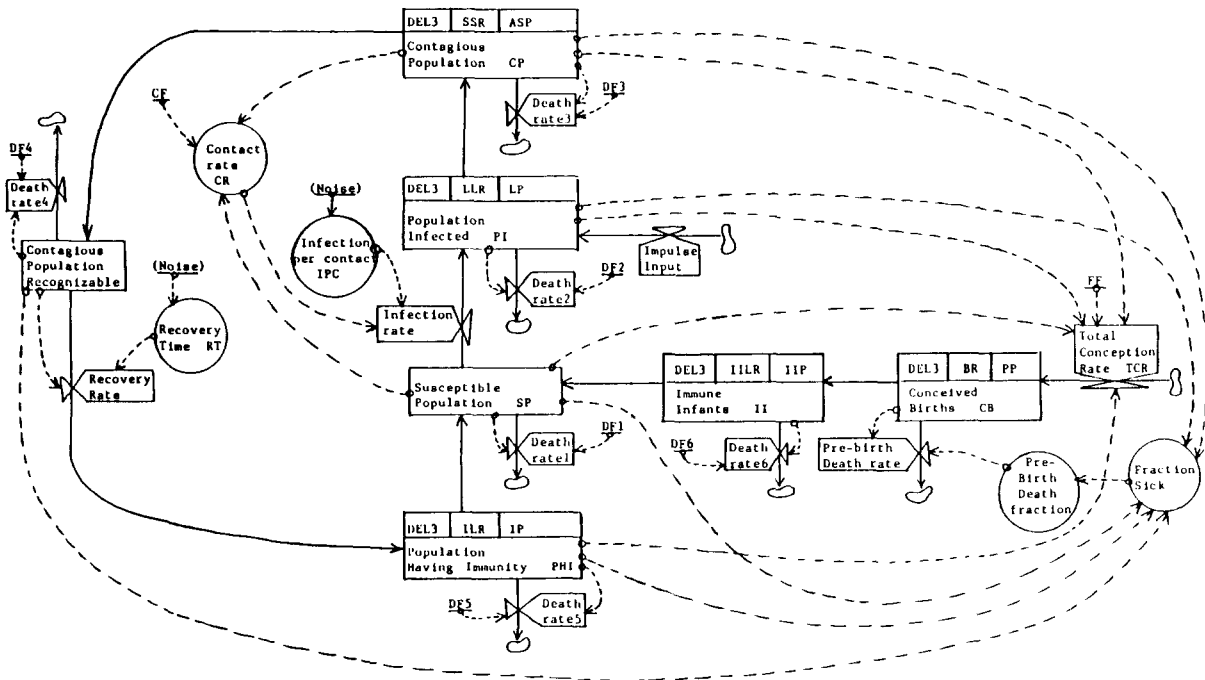


Figure 2. The flow diagram for the epidemic dynamics (the 'synthetic system')

IPC is a normal random variable with mean 0.10 and standard deviation 0.005:

$$IPC = 0.10 + \text{SAMPLE}(\text{NOIS1}, \text{INT1}, 0.0),$$

$$\text{INT1} = 3.0 \text{ Sampling interval (months),}$$

$$\text{NOIS1} = \text{NORMRN}(0.0, \text{STDV1}) \text{ Normal RV,}$$

$$\text{STDV1} = 0.005 \text{ Standard Deviation}$$

$$(5\% \text{ of the average}).$$

SAMPLE and NORMRN are built-in DY-NAMO functions. NORMRN generates random variates that are normally distributed with a given mean and standard deviation. When NORMRN is used in an equation, it generates an independent random value at *every* simulation step (DT). But this sort of noise generation is inappropriate for typical SD-type of systems for two reasons. First, in such systems most noise forces do *not* exhibit purely random (independent) fluctuation every hour, day or even week. Instead, in such feedback systems (with large time constants), the major random changes will be every couple of weeks, months or even years. In other words, since successive short-term (hourly, daily, weekly) random changes are strongly correlated, the fundamental period of random variation will be several weeks or months. Second, even if such feedback systems did have few noise components that have high frequency variations, these would have negligible effects on the broad patterns of behavior generated by the interactions of feedback loops with large time delays. (See Forrester [8] Appendix F, Peterson [14], Richardson and Pugh [15], and Eberlein and Wang [7] for further discussion of estimation of noise characteristics.) For these reasons, the SAMPLE function is used to assure that the value of IPC changes every INT1 time unit, rather than at every simulation step. (In the base run, INT1 = 3 months. In other runs, we will experiment with different levels of sampling intervals for different noise components.)

The 'contact rate' CR is given by

$$CR = CF * SP * CP \text{ contacts per month,}$$

$$CF = 0.004 \text{ Contact Fraction (per month),}$$

$$SP: \text{ Susceptible Population,}$$

$$CP: \text{ Contagious Population}$$

$$(SP * CP \text{ gives 'all possible contacts'}).$$

The second noisy equation is the 'recovery rate' RR:

$$RR = \text{CPR} / \text{RT people/month,}$$

$$\text{CPR: Contagious Population Recognizable.}$$

RT is the 'Recovery Time', chosen to be a normal random variable with mean 2 months and standard deviation 0.2:

$$RT = \text{RTAVG} + \text{SAMPLE}(\text{NOIS2}, \text{INT2}, 0.0),$$

$$\text{RTAVG} = 2,$$

$$\text{INT2} = 0.5 \text{ sampling interval,}$$

$$\text{NOIS2} = \text{NORMRN}(0.0, \text{STDV2}) \text{ Normal RV,}$$

$$\text{STDV2} = 0.10 * \text{RTAVG} (= 0.2).$$

Once again, the SAMPLE function is used to assure that RT changes every 0.5 months (INT2 = 0.5), rather than at every simulation step.

Finally, we assume that the *observed variable* 'Contagious Population Recognizable' CPR is distorted with 'observation error' ('measurement error'):

$$\text{OBS} = \text{CPR} + \text{SAMPLE}(\text{ONOIS}, \text{OINT}, 0.0),$$

$$\text{OINT} = 1.0 \text{ Sampling interval}$$

$$\text{for observation error,}$$

$$\text{ONOIS} = \text{NORMRN}(0.0, \text{OSTDV}) \text{ Normal RV,}$$

$$\text{OSTDV} = 0.05 * \text{CPR.}$$

In this particular illustration, the observation error is a Normal variable that can change at every observation interval (which is 1 month), and has a standard deviation equal to 5% of the actual value of CPR.

The major structures used in the Synthetic system are well known in mathematical epidemiology, but the parameters values are fictitious. What is important for this project is that the synthetic system has some generalizable *realistic* structures and parameter values: *Non-linear* relationships, coupling of multiple *positive* and *negative* feedback loops, inclusion of *system noise*, inclusion of *observation error* and a system order high enough (5 third-order and 2 first-order level equations, adding to 17) to create a realistic internal momentum. Another important characteristic of the system is its *generalizable structure*: The basic structure of the epidemic model, with minor modifications, can be observed in models that represent the dynamics of the spread of information

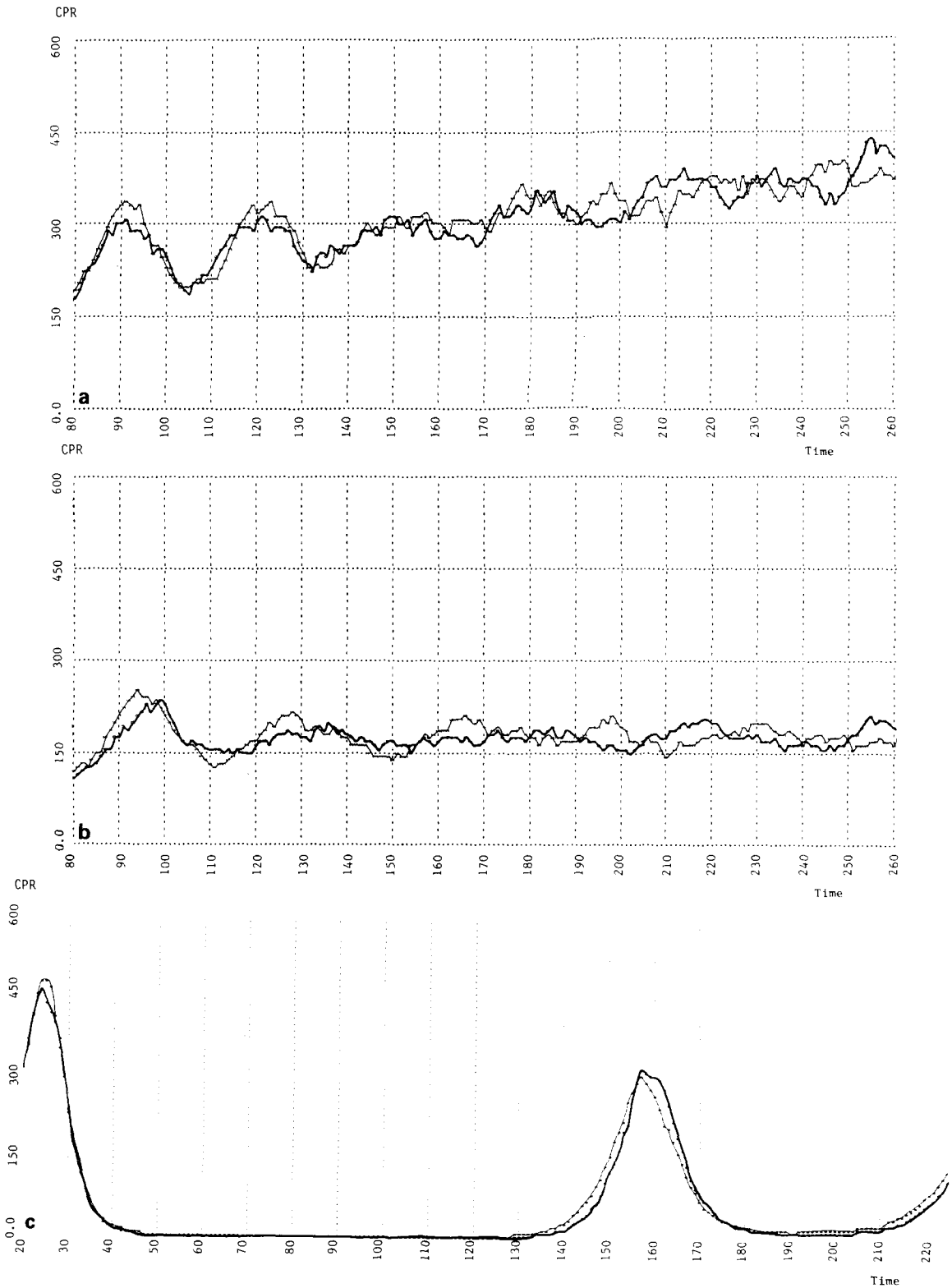


Figure 3. Three types of behavior patterns exhibited by the synthetic system under three different conditions. (Each type is illustrated by two different noise seeds, represented by the thick line and the thin line.) (a) Oscillations with population growth. (b) Oscillations without population growth. (c) 'Recurrent epidemics' behavior when immunity is permanent

among people, the spread of technological innovations (marketing) and certain predator–prey dynamics. Finally, an important property of the synthetic system is that it allows us to experiment with *three fundamentally different types* of behavior patterns: When both the immunity losing loop and the population growth loops are active, the system exhibits the oscillatory growth pattern seen in Figure 3(a). When the growth loops are made inactive, we observe constant-mean oscillations (Figure 3(b)), and when the immunity is assumed to be *permanent*, then the behavior changes to the ‘recurrent epidemic’ type seen in Figure 3(c).

Contrasting the three types of behavior patterns of Figure 3(a), 3(b), and 3(c), we see that there is a fundamental difference between the behavior of 3(c) on the one hand, and the behavior patterns of 3(a) and 3(b) on the other hand. The ‘recurrent epidemic’ behavior seen in Figure 3(c) is highly system-specific, *non-stationary* and *transient*. Basic time-series concepts such as means, variances, trend slopes and periods become meaningless when applied to such behavior patterns. Such highly transient behavior can *not* be characterized by its ‘mean’, ‘variance’, ‘slope’, or ‘period’. These standard measures can *not* capture the defining characteristics of the behavior pattern of Figure 3(c). Therefore, the suggested multi-step validation procedure is not suitable for this type of highly transient, non-stationary behavior pattern. In the experiments that follow, the performance of the multi-step procedure will be tested only on the ‘growth’ and ‘no-growth’ patterns seen in Figures 3(a) and 3(b).

As for the ‘recurrent epidemic’ type of behavior of Figure 3(c), a much more appropriate validation procedure is to use *graphical* measures of the most significant characteristics of such behavior patterns. For instance, one would graphically measure the time intervals between epidemic breakouts, how these intervals vary (get shorter) through time, how the magnitudes of the epidemics decrease through time, how long each epidemic lasts, etc. The behavior pattern validation would consist of measuring these non-stationary characteristics of both the model and the real behavior and then compare them to see if the model is able to generate these major characteristics.

6. The results

To generate the synthetic ‘real’ (‘observed’) output, the noise seed is set to 7654321. To generate the ‘model outputs’, for every experimental design, the simulation is run four times with four different noise seeds (6654321, 5654321, 4654321, 3654321). Then, the ‘real output’ is compared to the four model outputs (replications) by using the quantitative procedures described in Section 3. The results from the different simulation experiments are summarized in Table 1. A FORTRAN program is written to do the computations and the plotting. (The listing of the program is given in Appendix 2.) The external library functions FTAUTO and FTCROS are called from the IMSL scientific library software. All the computations are based on the segments of the outputs from $t = 50$ to $t = 300$. The program first fits a linear regression line to the ‘real’ and ‘model-generated’ outputs and detrends them if the analyst wishes to. Next, the autocorrelation function, the partial autocorrelation function and their standard errors (Se), the crosscorrelation function, the ‘percent error in the means’, the ‘percent error in the variations’ and the ‘discrepancy coefficient’ U are computed and printed out. The program also plots the autocorrelation functions of the ‘real’ and ‘model-generated’ outputs, the difference between the two functions (together with $2Se$ bands), and the autocorrelations of the residual series $e_i = S_i - A_i$.

6.1. The effect of system noise only

As the first set of experiments (the first experimental row of Table 1), we take the idealized situation where the model not only has a perfectly accurate structure, but it is also free from input function errors, parameter errors and observation errors. Thus, the *only* source of discrepancy is the difference between the noise sequences used in the synthetic real system and in the model. (This is achieved by changing the noise seed.)

We consider first the effect of ‘moderate’ system noise: $STDV1 = 0.005$, $INT1 = 3$, $STDV = 0.2$, $INT2 = 0.5$. The real system behavior (thick line) and the model behavior corresponding to seed 6654321 (thin line) for this case are shown in Figure 3.

Table 1
Summary results from various simulation experiments. Each experimental cell has four replication runs

The experimental condition	Significant difference in			Maximum cross-correlation and its lag	Percent error in		Discrepancy coefficient U					
	Trends	Autocorr. functions	Means E_1		Variations E_2							
Growth	Moderate	No	No	No	0.738 at 0	0.647 at 0	0.028	0.008	0.091	0.004	0.364	0.420
	system noise	No	No	No	0.802 at 2	0.794 at -1	0.008	0.005	0.186	0.206	0.354	0.353
	Increased	No	No	No	0.443 at 1	0.268 at 1	0.028	0.022	0.167	0.058	0.534	0.608
	system noise	No	No	No	0.610 at 3	0.660 at -2	0.007	0.034	0.286	0.333	0.511	0.466
	Independent ob-	No	No	Crit.	0.706 at 0	0.611 at 1	0.032	0.015	0.012	0.076	0.384	0.446
	servation errors	No	No	Crit.	0.757 at 2	0.762 at -2	0.023	0.015	0.254	0.337	0.398	0.405
	Correlated obser-	No	No	No	0.727 at 1	0.634 at 1	0.032	0.028	0.038	0.044	0.374	0.431
	vation errors	No	No	No	0.758 at 2	0.747 at -2	0.023	0.016	0.246	0.372	0.389	0.417
	'Hi-gain' pa-	No	Crit.	No	0.803 at 1	0.758 at 1	0.190	0.163	0.631	0.637	0.390	0.421
	rameter error	Crit.	Crit.	Crit.	0.827 at 2	0.820 at -1	0.145	0.164	0.944	0.906	0.444	0.425
No growth	'Lo-gain' pa-	Yes	Yes	Crit.	0.462 at 13	0.362 at 13	0.248	0.276	0.353	0.015	0.821	0.803
	rameter error	Yes	Yes	Crit.	0.512 at 11	0.276 at 7	0.286	0.264	0.029	0.048	0.769	0.694
	Moderate	-	No	No	0.742 at 0	0.768 at 0	0.012	0.006	0.045	0.005	0.360	0.341
	system noise	-	No	No	0.870 at 0	0.814 at -2	0.004	0.010	0.261	0.108	0.278	0.347
	Increased	-	No	No	0.395 at 0	0.449 at 1	0.019	0.020	0.130	0.022	0.553	0.530
	system noise	-	No	No	0.727 at 1	0.656 at -3	0.005	0.012	0.340	0.222	0.405	0.493
	Independent ob-	-	Crit.	Yes	0.733 at 0	0.725 at 1	0.008	0.006	0.008	0.036	0.365	0.375
	servation errors	-	No	Crit.	0.839 at 1	0.789 at -2	0.0002	0.009	0.271	0.182	0.308	0.383
	Correlated obser-	-	No	No	0.730 at 0	0.737 at 1	0.016	0.004	0.048	0.020	0.368	0.363
	vation errors	-	No	No	0.828 at 0	0.785 at -2	0.002	0.010	0.293	0.197	0.318	0.377
'Hi-gain' pa-	rameter error	-	No	No	0.775 at 1	0.775 at 1	0.163	0.155	0.992	1.05	0.459	0.477
		-	No	No	0.867 at 1	0.841 at 0	0.156	0.160	1.28	1.03	0.468	0.431
'Lo-gain' pa-	rameter error	-	Crit.	Yes	0.428 at -22	0.456 at 16	0.272	0.278	0.308	0.037	0.852	0.849
		-	No	Crit.	0.622 at 13	0.389 at 14	0.284	0.272	0.184	0.146	0.822	0.779

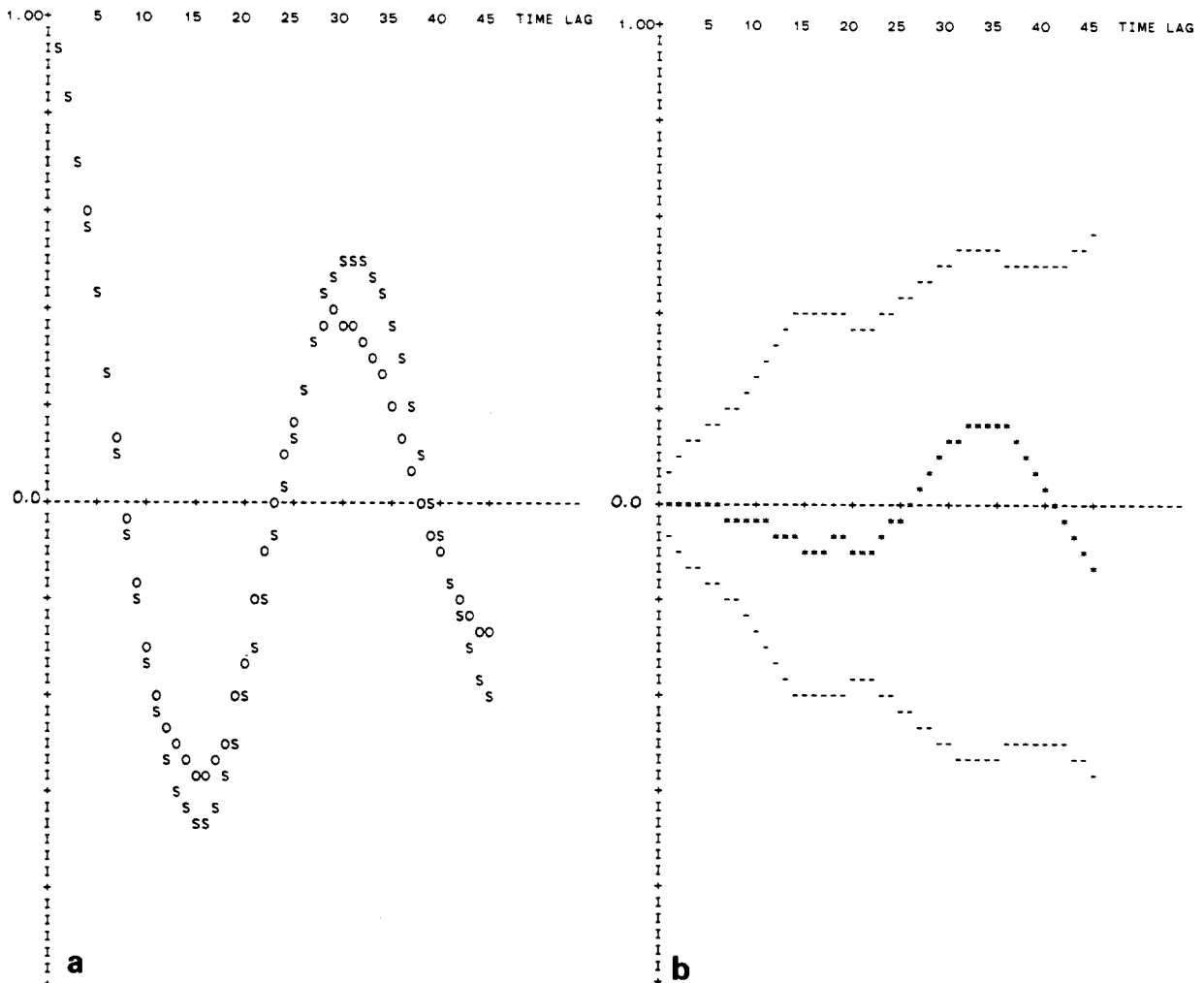


Figure 4. (a) Autocorrelation functions of the time patterns of Figure 3(a) (after trend removal). The Symbol 'o' stands for 'observed' and 's' for 'simulated'. (b) Differences of the two autocorrelation functions of Figure 4(a), and the 2Se confidence band

We will illustrate the application of the suggested validation procedures on the behavior patterns of Figure 3(a) (the 'growth case'). The autocorrelation functions (of the *detrended* patterns) are plotted in Figure 4(a). Note that the autocorrelation functions estimate the major periods (≈ 32) of the corresponding time patterns. In Figure 4(b), we see that the two functions are not significantly different at any lag. The crosscorrelation function (shown in Table 2) reaches a maximum value of 0.738 at lag 0 (meaning *no* phase lag between the two behavior patterns).

Next, we momentarily diverge from the six-step validation procedure and plot in Figure 5, the autocorrelation function of the *residual series* $e_i = S_i - A_i$. Our purpose here is to demonstrate the

absurdity of requiring accurate *point forecasts* from SD-type models. The autocorrelation of residuals (Figure 5) shows that the residuals are strongly autocorrelated and even cyclic, in spite of the fact that the model structure and the real system structure are 100% the same (except for the noise sequences used in the two, which must differ by the definition of noise)! This is a natural result of the fact that SD models generate strongly autocorrelated behavior patterns. Thus, even when the *same* model is run with two different noise sequences, it will generate two behavior patterns with consistent and sustained (autocorrelated) errors on a point-by-point basis. (See Forrester [8], Appendix K, and Wright [20] for further demonstration of this.) We must therefore test the *pat-*

Table 2

The cross-correlation function between the 'observed' and 'simulated' behavior patterns of Figure 3(a) (after trend removal)

Lag	Correlation
-25	0.1085
-24	0.0209
-23	-0.0641
-22	-0.1579
-21	-0.2427
-20	-0.3215
-19	-0.3941
-18	-0.4556
-17	-0.4959
-16	-0.5104
-15	-0.4997
-14	-0.4704
-13	-0.4283
-12	-0.3722
-11	-0.3024
-10	-0.2140
-9	-0.1126
-8	0.0012
-7	0.1213
-6	0.2434
-5	0.3549
-4	0.4512
-3	0.5374
-2	0.6272
-1	0.6965
0	0.7383
1	0.7296
2	0.6952
3	0.6299
4	0.5454
5	0.4405
6	0.3210
7	0.2023
8	0.0691
9	-0.0805
10	-0.2179
11	-0.3260
12	-0.4186
13	-0.4807
14	-0.5357
15	-0.5672
16	-0.5813
17	-0.5744
18	-0.5540
19	-0.5074
20	-0.4425
21	-0.3622
22	-0.2684
23	-0.1594
24	-0.0537
25	0.0570

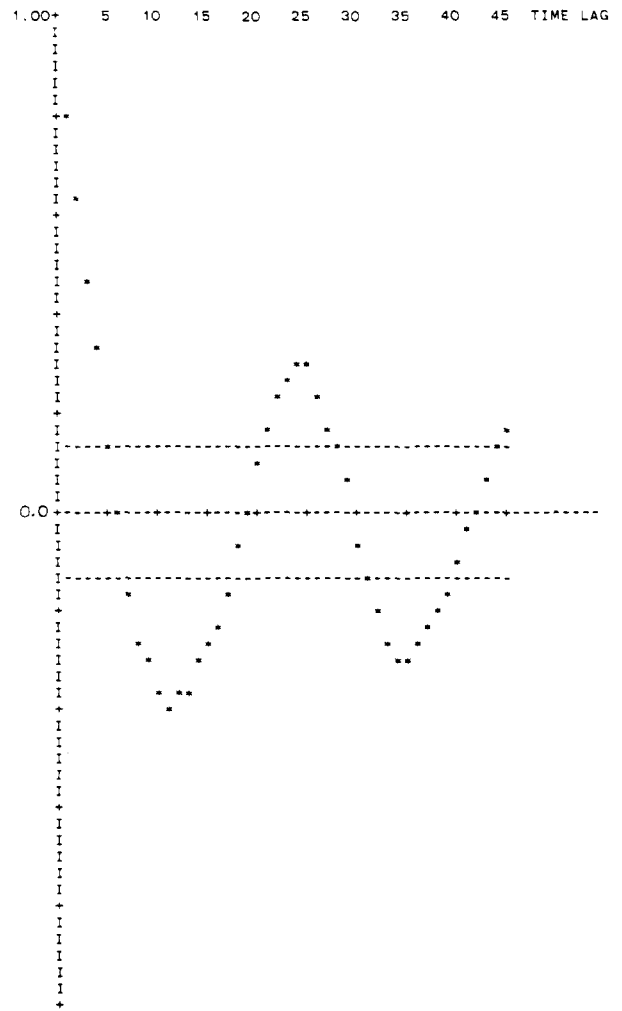


Figure 5. Autocorrelation function of the residuals $S_t - A_t$, and the 2Se confidence band

Table 3

Summary statistics for the behavior patterns of Figure 3(a) (after trend removal)

Regression coefficients for observed trend:

$$B0 = 212.6 \text{ Var}(B0) = 22.70 \quad B1 = 0.818 \text{ Var}(B1) = 0.0011$$

Regression coefficients for simulated trend:

$$B0 = 218.5 \text{ Var}(B0) = 18.73 \quad B1 = 0.785 \text{ Var}(B1) = 0.0009$$

Summary statistics:

Observed:	M1 = 212.6	Simulated:	M1 = 218.5
	M2 = 37.4		M2 = 34.0
	M3 = 30.3		M3 = 22.0
	M4 = 53.1		M4 = 45.8

Percent moment errors:

$$E1 = 0.0280$$

$$E2 = 0.0915$$

$$E3 = 0.2717$$

$$E4 = 0.1368$$

Discrepancy coefficient: $U = 0.3645$

tern prediction ability of SD models (as supported in this case by the autocorrelation function of Figure 4(a) and the crosscorrelations of Table 2).

In Table 3, we see that slope estimates for the real and model-generated patterns are 0.818 and 0.785 respectively, which are not significantly different. The 'percent error in the means' $E1 = 0.028$ and the 'percent error in the variations' $E2 = 0.0915$. Finally, note that the overall 'discrepancy coefficient' U can be as large as 36%, although the model has a *perfect* structure with moderate noise. (Once more, an illustration of the impossibility of generating highly accurate 'point forecasts' using SD models.)

We repeat the same experiments with the three other noise seeds (5654321, 4654321 and 3654321), the results of which are summarized in Table 1. We then change the experimental condition to the 'increased' system noise by doubling STDV1 (to 0.01) and INT2 (to 1.0). The summary results are shown across the second experimental condition in Table 1. When compared to the moderate noise setting, the statistics indicate poorer correspondence as expected: the crosscorrelations drop, $E1$, $E2$, and U increase. The trend estimates and the autocorrelation functions still indicate good agreement, meaning they were able to measure the major time patterns buried in increased noise.

Finally, the above experiments are repeated for the 'no-growth' case, with both 'moderate' and 'increased' noise conditions. The corresponding results are summarized across the first two experimental conditions of the second half of Table 1. The outcomes are quite similar to the ones obtained from the 'growth' case discussed above.

6.2. The effect of observation errors

Observation error (measurement error) is the difference between the observed (measured) value of a variable and its actual value. In this set of experiments, we add 5% observation error to the actual value of CPR, before its value is recorded. First we assume that the observation errors are *independent* from one observation period to the other. This is achieved by letting $OINT = 1$. As expected, the measures of correspondence deteriorate compared to the cases with no observation errors. (Compare the third and the first experimental conditions of Table 1.) Note how the inclusion of independent observation errors gets

reflected in the autocorrelation functions: In three of the four replications the autocorrelation functions exhibit 'critical' differences. (We define a 'critical' difference to mean a difference smaller than $2Se$ but larger than $1.5 Se$.) In all of those three runs, the differences occur within the first 3 to 4 lags. This is due to the fact that the independent (high-frequency) observation errors lower substantially the autocorrelations of the observed time patterns at small lags. In such cases, when we know that the small lag differences in the autocorrelation functions are caused by high-frequency observation errors (or any kind of high-frequency noise), these small lag differences can be ignored in judging the model's ability to predict the broad patterns of behavior. We can actually use the autocorrelation functions to find out whether the 'real' behavior pattern has high-frequency noise components not present in the model behavior pattern.

Next, we make the observation errors *correlated* by setting $OINT = 3$. The results obtained are shown across the fourth experimental cell of Table 1. In general, we observe that the correspondence measures are very similar to the ones obtained in the independent case. The only obvious change is that the autocorrelation tests do *not* yield 'critical' differences in this case. This is due to the fact that since the observation errors are now correlated, they do *not* have a lowering effect on the small lag autocorrelations of the observed time patterns.

When the above experiments are repeated for the 'no-growth' case, the results are similar to the 'growth' case discussed above. In both cases, we see that the effect of observation errors (whether independent or correlated) is quite small compared to the effect of *system noise*. We have seen in the previous experiments that, system noise, traveling through the feedback loop, can cause substantial behavior discrepancies, even pattern distortions such as irregular phase shifts and amplitude variations.

6.3. The effect of parameter errors

As the first set of parameter errors, we increase the parameter IP from 15 to 18, RTAVG from 2 to 2.4 and CF from 0.0040 to 0.0048 (all changes are 20%). The purpose of this 'hi-gain' setting is to increase the amplitudes of the oscillations of the model output. A secondary effect is to raise slightly

the average value of the oscillations. These effects are observed in the E2 and E1 values across the fifth experimental cell of Table 1.

As the second set of parameter errors, IP is again increased to 18, but CF is *decreased* by 20% (to 0.0032). The purpose of this 'low-gain' setting is to lengthen the periods of the oscillations and

lower the mean values. The period errors are detected by the autocorrelation functions and the mean errors by the E1 values. Note also that the crosscorrelation functions have rather small maxima and they occur at very large lags (like +13 and -22). *But* these lags do *not* point to a phase lag between the observed and model-generated

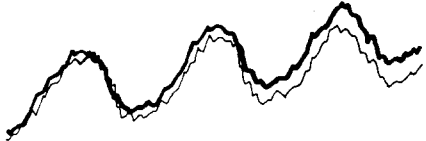
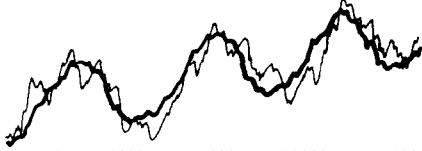
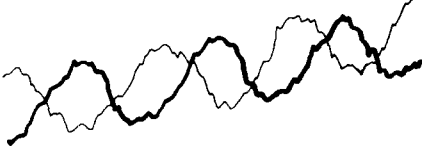
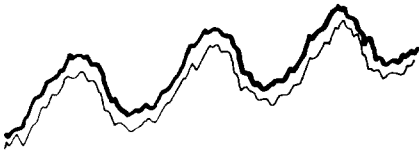
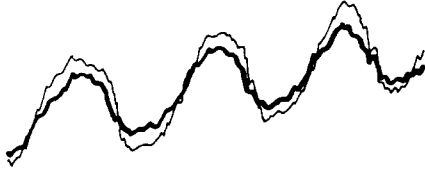
TYPE OF DISCREPANCY (EXAGGERATED FOR VISUAL CLARITY)	DISCREPANCY INDICATED BY:				
	COMPARISON OF TREND SLOPES	COMPARISON OF AUTOCORR. FUNCTIONS	THE LAG OF THE MAX CROSSCORR. FUNCTION	% ERROR IN THE MEANS	% ERROR IN THE VARIATIONS
	(Step 1)	(Step 2)	(Step 3)	(Step 4)	(Step 5)
	No	Yes	No	No	Yes
	Yes	No	No	Not Inter- pretable	No
	No	Yes	Not Interpret- able	No	No
	No	Yes	No	No	Yes
	No	No	Yes	No	No
	No	No	No	Yes	No
	No	No	No	No	Yes

Figure 6. The effects of some major pattern discrepancies between the observed behavior (thick line) and the model behavior (thin line) on the suggested behavior pattern measures

behavior patterns. Rather, they are mere artifacts of the *period errors* between the two behavior patterns. Therefore, if a significant period error is present, the crosscorrelation function does *not* have a clear interpretation.

Finally, a secondary effect of the 'low-gain' setting is an increase in the slope of the corresponding behavior pattern. In the related entries of Table 1, the trend estimates show significant differences. At this point, it is important to note that whenever there is a significant difference in the trend components, comparing the *mean values* of the behavior patterns becomes meaningless. When behavior patterns involve trends, the 'percent error in the mean' E1 is interpretable only if the trend components are *not* significantly different.

6.4. The effect of input error

The Synthetic System is excited at $t = 8$ with an impulse function, creating a sudden inflow of 100 infected people. To produce a phase lag between the observed and model-generated behavior patterns, we change the excitation time of the model to $t = 18$ (a lag of 10 months). When the resulting behavior is compared to the observed behavior, we see that the crosscorrelation function reaches its maximum at around lag +10 in all four replications: For the 'growth case', the estimates at 9, 9, 10 and 7; for the 'no-growth case', they are 9, 10, 11 and 9. We see that the cross-correlation function provides adequate estimates of the phase lag. But the cross-correlation function becomes problematic when system noise is large enough to result in irregular phase shifts, and hence causes the function to have its maximum at a non-zero lag. Two criteria are important in deciding whether a phase lag indicated by the cross-correlation function is *random or systematic*. *First*, when the phase error is random, the cross-correlation function becomes maximum sometimes at positive lags and sometimes at negative lags in a random fashion, as the noise seed is changed. But when the phase error is *systematic*, the crosscorrelation function indicates *consistently* a positive (or negative) lag, regardless of the noise seed. *Second*, in case of a systematic phase lag, the maximum and the minimum of the crosscorrelation function are relatively large compared to the case of a random phase lag. One may check the quantity $|\max -$

$\min|$ and suspect a systematic phase shift if the quantity is 'large enough'. (In our experiments, in the presence of systematic phase shift, $|\max - \min|$ was always larger than 0.80.)

The experiments of this section show that each individual step of the suggested validation procedure focuses on an individual pattern error: Trend errors are detected by comparing the slope coefficients, period errors by comparing the autocorrelation functions, phase errors by the lag at which the crosscorrelations become maximum, mean errors by large values of E1 and variation (or amplitude) errors by large values of E2. We provide a 'visual version' of these results in Figure 6. Each row depicts a typical individual pattern discrepancy, and the columns indicate which steps of the procedure detect that discrepancy. It is important to observe that in some instances the outcome of a specific step becomes *uninterpretable* if the behavior has failed in some previous steps. Due to these inter-dependencies, the multi-step procedure must be implemented in the given *logical sequence* in order to be most informative. Note also that the different 'composite' errors (such as *both* period, trend and amplitude errors in the *same* behavior) are not included in Figure 6, simply because all possible combinations would have unnecessarily made the Figure exceedingly large. To extend the given results to any composite pattern discrepancy is a straightforward process.

7. Experiments with a different synthetic system

In the previous section, simulation experiments were carried out on a 'synthetic system' in order to test the performance of the multi-step behavior validation procedure. As an extension, in this section we repeat the experiments with a different 'synthetic real system'. We choose Jay Forrester's Market Growth Model¹ as the new Synthetic System and test the performance of our multi-step validation procedure.

¹ This model was first published in "Market growth as influenced by capital investment", by Jay W. Forrester, *Sloan Management Review* 9/2, 83-105. The model equations are partly reprinted in Appendix 3 by permission of the Publisher. Copyright © 1968 by Sloan Management Review Association. All rights reserved.

For a complete description of the structure of the Market Growth Model the reader is referred to Forrester [10]. Here, we only note those parts of the model modified in this research. (The complete list of model equations is given in Appendix 3.) There are three minor modifications:

1. The constant TDDRC from 4 to 2.

2. The constant SEM from 400 to 800.

3. The table function TCEF from
 $-0.07/-0.02/0/0.02/0.07/0.15$

to

$-0.03/-0.01/0/0.02/0.07/0.15$.

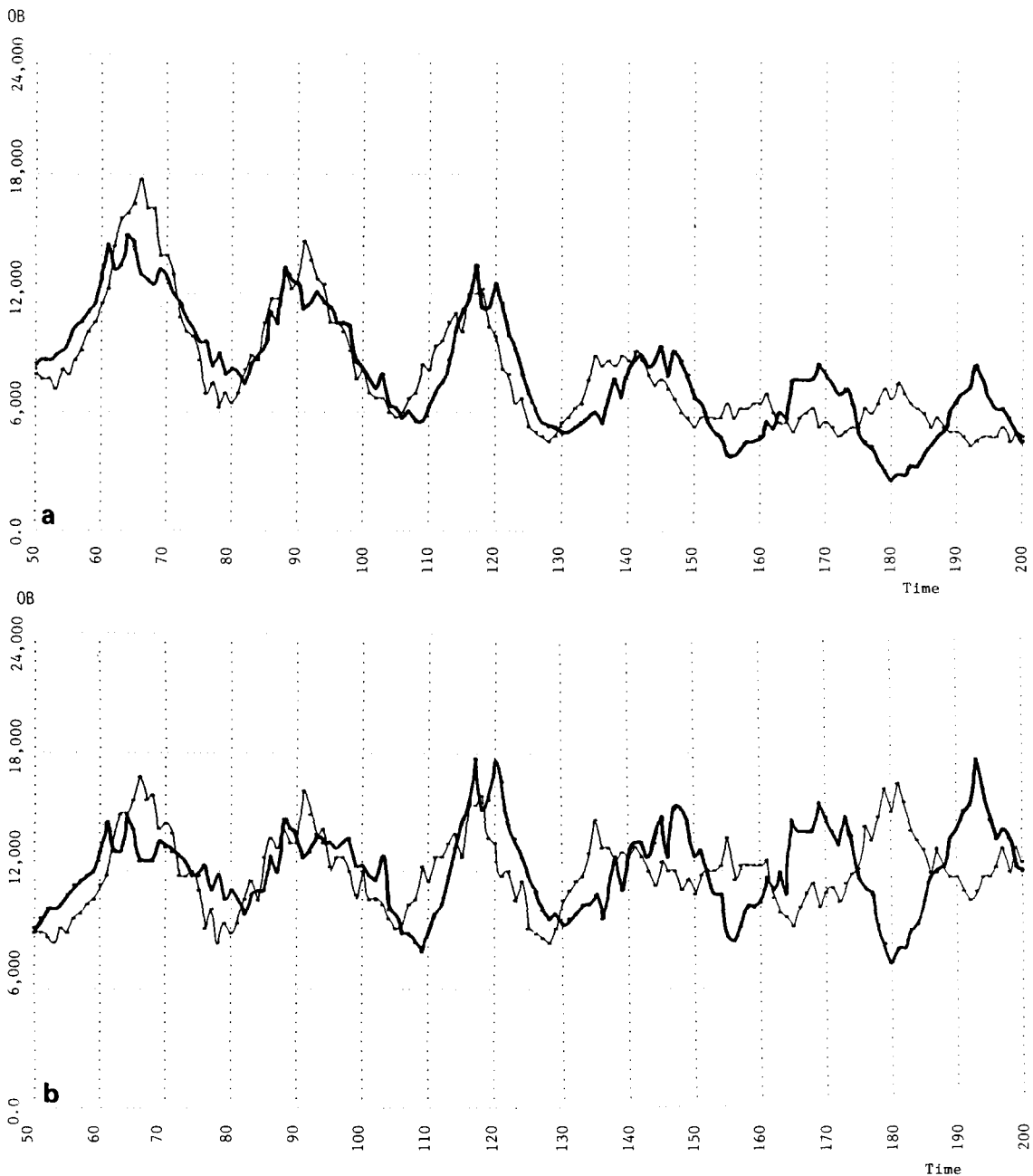


Figure 7. Two types of behavior patterns exhibited by the market growth model under two different conditions. (a) Variable production capacity—Oscillatory decline. (b) Constant production capacity—Oscillations with a constant mean

Table 4
Description of the experimental conditions for the market growth model

Experimental condition:	Corresponding parameter values					
	INT1	S1	INT2	S2	OINT	OFRAC
Moderate system noise	2.0	5%	1.0	5%	–	0%
Increased system noise	3.0	5%	2.0	10%	–	0%
Independent observation errors	2.0	5%	1.0	5%	1.0	5%
Correlated observation errors	2.0	5%	1.0	5%	3.0	5%

For our purposes, the significant effect of these changes is to create an output behavior with stronger oscillations and shorter periods. (Period was about 35 in the original model, it is about 25 in our version.) The importance of experimenting with this type of behavior is that it is much closer to the border of ‘unstability’ than the behavior of the epidemic model. The only other modification is the inclusion of noise factors. This is necessary by the very nature of our experimental study. (The original Market Model is entirely deterministic.) System noise is introduced in two different locations. The related equations are marked in the model listing given in Appendix 3.

Finally, the ‘observed’ variable is assumed to be the Orders Booked (OB). An observation error is introduced to the observed value, in the same manner as it was done for the Epidemic Model in the previous section. The Orders Booked can exhibit two different types of behavior pattern depending on whether the Production Capacity is fixed or variable. When the Production Capacity is variable, the Orders Booked exhibits an *oscillatory decline* pattern (as illustrated in Figure 7(a) with two different random number seeds). When the Production Capacity is fixed at 12000 units, the behavior is oscillatory around a *constant-mean* (as illustrated in Figure 7(b)). Simulation experiments are carried out for both cases and the multi-step validation procedure is applied. In all experiments, the ‘real’ behavior is generated with seed 9999999 and model behavior patterns are generated with seeds 7777777, 5555555, 3333333 and 1111111. In this section, we will not describe and discuss all these experiments. Instead, we will

give a summary of experimental conditions and a summary of results. The experimental conditions are summarized in Table 4. When these four experiments are run for the ‘oscillatory-decline’ and the ‘constant-mean’ cases, the results shown in Table 5 are obtained.

Comparing Table 5 to Table 1, we observe that in general, measures of discrepancy have *larger* values and measures of correspondence have *smaller* values. That is, for the models of the second synthetic system, our multi-step procedure in general indicates poorer behavior validity. This result should indeed be expected, because the new synthetic system being closer to unstability than the first one, its behavior is more sensitive to the variations in its noise parameters. Thus, from one noise sequence to another, we can get large discrepancies in the behavior patterns. This leads to an observation: We should not expect the same test results from models of different real systems. Models of ‘almost unstable’ systems would naturally produce poorer measures of correspondence than models of stable systems, other things being equal.

Our earlier remarks and observations about the *qualitative* properties of the multi-step procedure are *not* affected by the results of the second set of experiments. The suggested sequence of testing, what pattern components are measured by which tools; our remarks about their interrelationships, potential pitfalls involved in using the tools, etc. still hold. Furthermore, even the numerical values show some uniformity. When we consider the *average values* of four replications (rather than the individual replications), we see that E1 does not

Table 5
Summary of experimental results for the market growth model

The experimental condition	Significant difference in			Maximum cross-correlation and its lag	Percent error in		Discrepancy coefficient U					
	Trends	Autocorr. functions	Means E1		Variation E2							
Decline	Moderate system noise	No	No	No	0.660 at -1	0.814 at -3	0.006	0.031	0.096	0.340	0.429	0.458
		No	No	No	0.749 at -3	0.642 at -4	0.055	0.004	0.449	0.037	0.459	0.586
	Increased system noise	Crit	No	Crit	0.255 at -12	0.742 at -5	0.104	0.004	0.04	0.131	0.791	0.598
		No	No	Crit	0.598 at 0	0.688 at -3	0.054	0.002	0.354	0.201	0.488	0.470
	Independent observation errors	No	No	No	0.634 at -1	0.816 at -3	0.010	0.035	0.097	0.344	0.453	0.456
		No	No	Crit	0.750 at -3	0.620 at -4	0.054	0.003	0.454	0.44	0.466	0.590
	Correlated observation errors	No	No	No	0.629 at -1	0.834 at -3	0.029	0.047	0.147	0.310	0.446	0.441
		No	No	Crit	0.751 at -3	0.625 at -4	0.050	0.011	0.524	0.093	0.479	0.585
No decline	Moderate system noise	-	-	No	0.457 at -29	0.691 at -6	0.010	0.003	0.099	0.240	0.661	0.643
		-	-	No	0.562 at -7	0.495 at -8	0.008	0.012	0.177	0.116	0.696	0.779
	Increased system noise	-	-	Crit	0.547 at 9	0.465 at -6	0.000	0.014	0.049	0.197	0.829	0.712
		-	Yes	Crit	0.406 at 4	0.514 at 24	0.002	0.002	0.398	0.028	0.637	0.531
	Independent observation errors	-	-	No	0.414 at -29	0.690 at -5	0.008	0.002	0.064	0.259	0.691	0.636
		-	-	No	0.555 at -7	0.478 at -8	0.007	0.011	0.207	0.081	0.701	0.779
	Correlated observation errors	-	-	No	0.374 at -29	0.701 at -5	0.001	0.007	0.017	0.219	0.680	0.627
		-	-	No	0.559 at -7	0.464 at -8	0.004	0.001	0.284	0.045	0.702	0.771

go beyond 5%, whereas E2 does become as large as 30%. U changes between 35% and 70%, and max. cross-correlation between 45% and 80%.

8. Conclusions

In this paper, we try to show that *multiple tests* (rather than a single test) are needed for SD model behavior evaluation. We suggest a specific six-step behavior validation procedure, and investigate its appropriateness and properties through a series of simulation experiments. Each step focuses on a specific pattern component. But since outcomes of the different steps are *not* entirely independent, the six-step procedure must be implemented in a specific sequence in order to be informative. Here is a brief summary of the suggested procedure:

Step 1—Trend comparison and removal. The trend components (if any) must be compared as the first step, because if the model behavior has a significant error in the trend component, then the ‘error in the means’ and the ‘error in the variances’ statistics lose their meaning. If a significant error in the trends is detected, then a model revision is called for. The trend components must be removed after making sure that no significant error remains.

Step 2—Comparing the periods. The suggested *autocorrelation function test* is able to detect significant errors in the periods. High-frequency noise components affect only the very low lag autocorrelations. Hence, the test can also be used to find out if one behavior pattern has high-frequency components not present in the other. The test is quite insensitive to other types of pattern errors.

Step 3—Comparing the means. The ‘Percent Error in the Means’ E1 has rather small variability. When the model has no systematic error, E1 very rarely goes beyond 5%. *Note* that if there is a *significant difference in the trend* components (i.e. Step 1 not passed), then E1 *loses its meaning*.

Step 4—Comparing the variations. The ‘Percent Error in the Variations’ E2 has relatively large variability. Even if the model has no systematic error, E2 can be as large as 30%.

Step 5—Testing for phase lag. The *cross-correlation function* provides an estimate of a potential phase lag. *But*, the crosscorrelation function being maximum at a non-zero lag does not automati-

cally indicate a systematic phase lag, because system noise and/or observation errors may also cause unsystematic phase shifts. A systematic phase error must be suspected if the lag at which the crosscorrelation becomes maximum does *not* assume positive and negative values randomly, from one noise seed to another. Again, the maximum and minimum crosscorrelations are relatively large in case of systematic phase error. (In the experiments, the quantity $|\max - \min|$ was always larger than 0.80 in presence of a systematic phase lag.) Finally, the crosscorrelation function becomes ambiguous if there is a significant *error in the periods* of the referent time patterns (i.e. Step 2 not passed), because period errors strongly affect the shape of the crosscorrelation function.

*Step 6—*As a final step, only after all validity tests have been passed, the discrepancy coefficient U can be computed as a single summary measure. Since U is basically a point-oriented measure whereas SD models are pattern oriented, rather large U values must be tolerable in model evaluation. Experiments show that, even for models with no systematic errors, U can be as large as 70%. This result is in agreement with Rowland and Holmes [16] who analyze Theil’s coefficient in the context of dynamic mathematical models and suggest that values between 0.4 and 0.7 should imply average-to-good models.

Experiments also show that the numerical outcomes of the various tests depend on the nature of the *system* and type of pattern involved. Models of highly stable systems should be expected to yield more accurate behavior reproduction than models of (‘almost’) unstable systems. In general, behavior accuracy exhibits substantial variation from one noise seed to another. Therefore, the multiple tests suggested above should always be carried out by running the model several times with different noise seeds. Much of the variation is caused by *system noise*. The effect of observation error is quite weak compared to that of system noise, which, by travelling through the feedback loops, can cause significant behavior distortions such as irregular phase shifts and amplitude variations.

9. Limitations and extensions

Although simulation experiments suggest the appropriateness of the six-step behavior validation

procedure, the latter is not without limitations. As discussed in Section 5, the procedure is not suitable to evaluate highly non-stationary, transient behavior patterns. Statistical summary measures (such as the ones suggested) can not capture the defining characteristics of short, transient, non-stationary behavior patterns. To evaluate such behavior, the most appropriate approach is to use *graphical* measures of the major pattern characteristics. The suggested multi-step procedure is applicable to relatively long, oscillatory behavior patterns. Another limitation of the method is that it ideally assumes the autocorrelation and variance structure of noise terms have been properly modeled. Unfortunately, this is a difficult model identification (estimation) problem. In practice, SD models rarely devote substantial time and effort to noise estimation. Forrester [8] Appendix F, Peterson [14], Richardson and Pugh [15], and Eberlein and Wang [7] discuss noise characteristics and issues involved in estimating them. Practical guidelines are needed as to under what conditions noise must be included, and how it must be estimated. Although it is technically possible to carry out the suggested tests on noise-free models as

well, results would be less reliable, since in this case each test would be run only on one replication.

Suggested procedure must be extensively tested on additional models with different behavior characteristics. It is anticipated that the validation procedure would prove to be more applicable to certain types of behavior and less applicable to others. A related issue would be the case of behavior evaluation with respect to multiple variables. In such cases, the most practical approach would be to apply the procedure to each variable independently, and conclude that the model has behavior validity if it passes the tests with respect to all the variables of interest.

Finally, passing the suggested behavior tests does not automatically imply model validity. As discussed in the Introduction, the *structural* validity testing is a prerequisite for behavior prediction testing. For the purpose of this particular research, we assumed that the model had already passed the structural validation phase. Given that this is the case, a favorable outcome of the suggested multi-step procedure does contribute significantly to the credibility of the model.

Appendix 1

```
* EPIDEMICS BY YAMAN BARLAS, GEORGIA TECH, 1985.
OPT CN,SL
OPT PLW=61
FNCTN TEST(5,1)
NOISE 7654321
L SP,K=SP.U+DT*(ILR.UK+ILR.UK-IR.UK-DR1.UK) SUSCEPTIBLE POPULATION
N SP=SPN
C SPN=3000
R ILR.KL=(1-DF5)*DELAYP(RR.UK,IP,PHI) IMMUNITY LOSING RATE
C IP=15 MONTHS IMMUNE PERIOD
C DF5=0.02
R ILR.KL=(1-DF6)*DELAYP(BR.UK,IIP,II) INFANT IMMUNITY LOSING RATE
C IIP=6 MONTHS INFANT IMMUNITY PERIOD
C DF6=0.01
A NOIS1.K=NORMRN(0.0,STDV1)
A IPC.K=0.10+SAMPLE(NOIS1.K,INT1,0.0) INFECTION PER CONTACT
C INT1=3.0 SAMPLING INTERVAL
C STDV1=0.005
R IR.KL=IPC.K*CR.K INFECTION RATE
A INP.K=PULSE(AMP,START,RPEAT) INITIAL INPUT OF INFECTED
C AMP=400
C START=8.
C RPEAT=500.
R IRTOT.KL=IPC.K*CR.K+INP.K TOTAL INFECTION RATE
N IRTOT=IR
R DR1.KL=DF1*SP.K DEATH RATE 1
C DF1=0.002 PER MONTH DEATH FRACTION 1
C LP=2 MONTH LATENCY PERIOD
R LLR.KL=(1-DF2)*DELAYP(IRTOT.UK,LP,PI) LATENCY LEAVING RATE
N LLR=LLRN
C LLRN=0
C DF2=0.02
C ASP=2 MONTH ASYMPTOMATIC PERIOD
R SSR.KL=(1-DF3)*DELAYP(LLR.UK,ASP,CP) SYMPTOMS SHOWING RATE
C DF3=0.03
A CR.K=CF.K*SP.K*CP.K CONTACT RATE
A CF.K=0.004 PER MONTH CONTACT FRACTION
L CPR.K=CPR.U+DT*(SSR.UK-RR.UK-DR4.UK) CONTAGIOUS POPULATION RECOGNIZ.
N CPR=CPRN
```

Appendix 1 (continued)

```

C CPRN=0
R RR.KL=CPR.K/RT.K RECOVERY RATE
A RT.K=RTAVG+SAMPLE(NOIS2.K,INT2,0,0) RECOVERY TIME
C RTAVG=2. AVERAGE TIME TO RECOVER
C INT2=0.5 SAMPLING INTERVAL
A NOIS2.K=NORMRN(0,0,STDV2.K)
A STDV2.K=0.10*RTAVG
R DR4.KL=DF4*CPR.K DEATH RATE 4
C DF4=0.003
R BR.KL=(1-PBDF.K)*DELAYP(TCR.JK,PP,CB) BIRTH RATE
C PP=9 MONTHS PREGNANCY PERIOD
R TCR.KL=FF*(CP.K+PI.K+SP.K+PHI.K) TOTAL CONCEPTION RATE
C FF=0.006 PER MONTH FERTILITY FRACTION
A PBDF.K=DFLL+(DFUL-DFLL)*FS.K PRE-BIRTH DEATH FRACTION
C DFLL=0.02
C DFUL=0.03 DEATH FRACTION UPPER LIMIT
A FS.K=(PI.K+CP.K+CPR.K)/(PI.K+CP.K+CPR.K+SP.K+PHI.K) FRACTION SICK
NOTE
A OBS.K=CPR.K+SAMPLE(ONDIS.K,OINT,0,0) OBSERVED CPR
C OINT=1.00
A ONDIS.K=NORMRN(0,0,OSTDV.K)
A OSTDV.K=DFRAC*CPR.K
C DFRAC=0.05
C TO=52
A DUMM.K=TEST(COUNT.K,OBS.K,TIME.K,LENGTH,TO) DUMMY VAR. TO CALL FORTRAN FUNCT.
NOTE
SPEC DT=0.25/LENGTH=300/PRTPER=2/PLTPER=1
PLOT OBS=O(0,600)/PHI=M/SP=S/II=I
PRINT OBS,PHI,SP,PI,II
RUN REAL

```

```

C ... THE FOLLOWING IS AN EXTERNAL FUNCTION TO BE CALLED BY THE DYNAMO
C ... MODEL IN ORDER TO PRINT THE VALUES OF THE VARIABLE 'OBS' IN THE
C ... OUTPUT FILE 'OFILE'.

```

```

C
C     FUNCTION TEST(COUNT,OBS,TIME,SLNGTH,TO)
C     DIMENSION X(351)
C     OPEN(60,FILE='OFILE')
C
C ... 'TEST' IS JUST A DUMMY VARIABLE:
C     TEST=1
C
C ... THE VARIABLE 'OBS' IS PRINTED STARTING AT TIME TO:
C     IT0=TO
C     IF(TIME.EQ.0.0)COUNT=TIME
C ... STORE THE VALUES OF 'OBS' AT EVERY 1 TIME UNIT:
C     IF(TIME.LT.COUNT) RETURN
C     COUNT=COUNT+1.0
C     II=INT(COUNT)
C     X(II)=OBS
C     IF(COUNT.LE.SLNGTH) RETURN
C     WRITE(60,10) (X(KK) , KK=IT0,II)
10  FORMAT(12F6.1)
C     RETURN
C     END

```

Appendix 2

```

C THIS PROGRAM COMPUTES A SET OF SELECTED QUANTITATIVE MEASURES
C FOR THE OBSERVED AND MODEL-GENERATED BEHAVIOR PATTERNS.
C
C BY YAMAN BARLAS, 1984, GEORGIA INSTITUTE OF TECHNOLOGY.
C
C     PROGRAM MAIN(INPUT,OUTPUT,TAPES=INPUT,TAPE6=OUTPUT)
C     DIMENSION A(351),S(351),ARD(351),SRD(351),VARARD(50),VARSRD(50)
C     +,AACV(351),SACV(351),APAC(60),SPAC(60),WAPEA(60),STDDIF(60),
C     +ERRR(351)
C     DIMENSION XY(702),NC(4),EMUSIG(4),ACV(1,1,31),AC(1,1,31)
C     COMMON GRAF(61,60)
C     DATA SYMBS,SYMBO,HOR,STAR/"S","O","-","*"/
C
C READ THE OPTION CARD:
C N: TOTAL NUMBER OF OBSERVATIONS

```

Appendix 2 (continued)

```

C M: NUMBER OF AUTOCORRELATIONS TO BE COMPUTED.
C NOPT=0, DO NOT COMPUTE THE RESIDUAL AUTOCORRELATIONS AND DO NOT
C PLOT THE DIFFERENCES OF THE AUTOCORRELATION FUNCTIONS.
C NOPT=1, DO ALL THE COMPUTATIONS AND PLOTTING,
C EXCEPT THE RESIDUAL AUTOCORRELATIONS.
C NOPT=2, DO ALL THE COMPUTATIONS AND PLOTTING.
C ID=0, TREND REMOVAL IS NOT REQUIRED.
C ID=1, TREND REMOVAL BY FIRST ORDER DIFFERENCING.
C ID=2, TREND REMOVAL BY SIMPLE LINEAR REGRESSION.
C ISHFT=L, SHIFT THE MODEL BEHAVIOR PATTERNS BY (+ OR -) L TIME UNITS.
C ( NORMALLY ISHFT=0).
C
C READ(5,10) N,M,NOPT,ID,ISHFT
10 FORMAT(2I3,2I2,I3)
C
C READ FIRST THE OBSERVED DATA POINTS, THEN THE MODEL-GENERATED ONES.
C READ(5,20) (A(I),I=1,N)
C READ(5,20) (S(I),I=1,N)
20 FORMAT(12F6.2)
C
C IF(ISHFT.EQ.0) GO TO 201
C IF(ISHFT.GT.0) THEN
C N=N-ISHFT
C DO 202 I=1,N
202 S(I)=S(I+ISHFT)
C ELSE
C ISHFT=-ISHFT
C N=N+ISHFT
C DO 203 I=1,N
203 A(I)=A(I+ISHFT)
201 ENDIF
C DO 21 I=1,351
C ARO(I)=0.
21 SRO(I)=0.
C N1=N-1
C
C AUTOCORRELATIONS ARE COMPUTED BY THE EXTERNAL IMSL ROUTINE 'FTAUTO':
C CALL FTAUTO(A,N,N1,50,7,ABAR,AVAR,AACV,ARO,APAC,WAREA)
C ASTD=SQRT(AVAR)
C CALL FTAUTO(S,N,N1,50,7,SBAR,SVAR,SACV,SRO,SPAC,WAREA)
C SSTD=SQRT(SVAR)
C
C CALL HEAD
C
C DO 100 K=1,M
100 WRITE(6,101) K,ARO(K),SRO(K),APAC(K),SPAC(K)
101 FORMAT(17X,I2,2F11.3,9X,2F11.3)
C
C COMPUTE THE VARIANCES OF THE SAMPLE AUTOCORRELATIONS:
C CALL VAUTO(N,N1,ARO,VARARO)
C CALL VAUTO(N,N1,SRO,VARSR0)
C
C WRITE(6,22)
22 FORMAT(1H1//////////21X,'VARIANCES OF AUTOCORRELATION ESTI
+MATES')
C WRITE(6,23)
23 FORMAT(/23X,'LAG',5X,'OBSERVED',5X,'SIMULATED'/)
C DO 24 K=1,M
24 WRITE(6,25) K,VARARO(K),VARSR0(K)
25 FORMAT(23X,I2,2F13.4)
C
C CALL CLEAR
C
C PLOT THE AUTOCORRELATION FUNCTIONS:
C YMAX=1.0
C CALL PLOTTR(ARO,YMAX,M,SYMB0)
C CALL PLOTTR(SRO,YMAX,M,SYMB5)
C WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
259 FORMAT(1H1////////13X,F4.2,5A1,I1,8I5,2X,'TIME LAG')
C DO 26 I=2,61
26 WRITE(6,27)(GRAF(I,J),J=1,55)
27 FORMAT(17X,60A1)
C CALL CLEAR
C
C PLOT THE PARTIAL AUTOCORRELATION FUNCTIONS:
C CALL PLOTTR(AC,C,YMAX,M,SYMB0)
C CALL PLOTTR(SC,AC,YMAX,M,SYMB5)
C WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
C DO 29 I=2,61
29 WRITE(6,27)(GRAF(I,J),J=1,55)
C IF (ID.EQ.0)GO TO 299
C IF (ID.EQ.2) GO TO 281
C
C TREND REMOVAL BY DIFFERENCING:
C ARO(N-1)=0.
C SRO(N-1)=0.
C N=N-1
C DO 292 J=1,N

```

Appendix 2 (continued)

```

      A(J)=A(J+1)-A(J)
292  S(J)=S(J+1)-S(J)
      GO TO 289
C
C   TREND REMOVAL BY SIMPLE LINEAR REGRESSION:
281  CALL REGRES(A,N,ABO,AB1,VARABO,VARAB1)
      CALL REGRES(S,N,SBO,SB1,VARSB0,VARSB1)
      WRITE(6,282)
282  FORMAT(1H1//////////18X,'REGRESSION COEFFICIENTS FOR ',
+ 'OBSERVED TREND :')
      WRITE(6,283) ABO,VARABO,AB1,VARAB1
283  FORMAT(/18X,'BO=' ,F7.1,3X,'VAR(BO)=' ,F8.2,3X,'B1=' ,F7.3,3X,
+ 'VAR(B1)=' ,F7.4)
      WRITE(6,284)
284  FORMAT(///18X,'REGRESSION COEFFICIENTS FOR SIMULATED TREND :')
      WRITE(6,283) SBO,VARSB0,SB1,VARSB1
289  N1=N-1
C
C   AUTOCORRELATIONS OF THE TREND-REMOVED BEHAVIOR PATTERNS:
      CALL FTAUTO(A,N,N1,50,7,ABAR,AVAR,AACV,ARO,APAC,WAREA)
      ASTD=SQRT(AVAR)
      CALL FTAUTO(S,N,N1,50,7,SBAR,SVAR,SACV,SRO,SPAC,WAREA)
      SSTD=SQRT(SVAR)
C
C   VARIANCES OF THE SAMPLE AUTOCORRELATIONS:
      CALL VAUTO(N,N1,ARO,VARARO)
      CALL VAUTO(N,N1,SRO,VARSR0)
      CALL HEAD
      DO 293 K=1,M
293  WRITE(6,101) K,ARO(K),SRO(K),APAC(K),SPAC(K)
      WRITE(6,22)
      WRITE(6,23)
      DO 296 K=1,M
296  WRITE(6,25) K,VARARO(K),VARSR0(K)
      CALL CLEAR
C
C   PLOT THE AUTOCORRELATIONS OF TREND-REMOVED BEHAVIOR PATTERNS:
      YMAX=1.0
      CALL PLOTTR(ARO,YMAX,M,SYMB0)
      CALL PLOTTR(SRO,YMAX,M,SYMB5)
      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 294 I=2,61
294  WRITE(6,27) (GRAF(I,J),J=1,55)
      CALL CLEAR
C
C   PLOT THE PARTIAL AUTOCORRELATIONS OF TREND-REMOVED PATTERNS:
      CALL PLOTTR(APAC,YMAX,M,SYMB0)
      CALL PLOTTR(SPAC,YMAX,M,SYMB5)
      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 295 I=2,61
295  WRITE(6,27) (GRAF(I,J),J=1,55)
299  NC(1)=N
      NC(2)=N
      NC(3)=1
      NC(4)=30
      EMUSIG(1)=ABAR
      EMUSIG(2)=SBAR
      EMUSIG(3)=AVAR
      EMUSIG(4)=SVAR
      DO 11 I=1,N
      XY(I)=A(I)
11  XY(I+N)=S(I)
C
C   CROSSCORRELATIONS ARE COMPUTED BY THE EXTERNAL IMSL ROUTINE 'FTCROSS':
      CALL FTCROSS(XY,NC,4,EMUSIG,ACV,1,1,AC,1,1,IER)
C   CROSSCORRELATIONS ARE COMPUTED FOR 25 TIME LAGS:
      NCROS=26
      DO 12 J=1,NCROS
      JJ=NCROS+1-J
12  WAREA(J)=AC(1,1,JJ)
      DO 13 I=1,N
      XY(I)=S(I)
13  XY(I+N)=A(I)
      EMUSIG(1)=SBAR
      EMUSIG(2)=ABAR
      EMUSIG(3)=SVAR
      EMUSIG(4)=AVAR
      CALL FTCROSS(XY,NC,4,EMUSIG,ACV,1,1,AC,1,1,IER)
      WRITE(6,14)
14  FORMAT(1H1//////////18X,'CROSSCORRELATIONS BETWEEN SIMUL
+ATED AND OBSERVED SERIES')
      WRITE(6,15)
15  FORMAT(/31X,'LAG',7X,'CORRELATION'/)
      DO 16 J=1,NCROS
      JJ=J-NCROS
16  WRITE(6,17) JJ,WAREA(J)
17  FORMAT(31X,13,8X,F7.4)

```


Appendix 2 (continued)

```

      DO 18 J=2,NCROS
      JJ=J-1
18  WRITE(6,17) JJ, AC(1,1,J)
C
C  COMPUTE THE FOUR SAMPLE MOMENTS:
      CALL MOMT(N,A,ABAR,AM3,AM4)
      CALL MOMT(N,S,SBAR,SM3,SM4)
      WRITE(6,36)
36  FORMAT(1H1//////////18X,'SUMMARY STATISTICS : ')
      WRITE(6,35) ABAR,ASTD,AM3,AM4
35  FORMAT(/18X,'OBSERVED : ',2X,'M1=',F7.1/30X,'M2=',F7.1/30X,
+ 'M3=',F7.1/30X,'M4=',F7.1)
      WRITE(6,37) SBAR,SSTD,SM3,SM4
37  FORMAT(/18X,'SIMULATED : ',2X,'M1=',F7.1/31X,'M2=',F7.1/31X,
+ 'M3=',F7.1/31X,'M4=',F7.1)
C
C  COMPUTE THE 'PERCENT MOMENT ERRORS':
      E1=ABS(SBAR-ABAR)/ABS(ABAR)
      E2=ABS(SSTD-ASTD)/ASTD
      E3=ABS(SM3-AM3)/ABS(AM3)
      E4=ABS(SM4-AM4)/AM4
      WRITE(6,52)
52  FORMAT(/18X,'PERCENT MOMENT ERRORS : ')
      WRITE(6,54) E1,E2,E3,E4
54  FORMAT(/31X,'E1=',F6.4/31X,'E2=',F6.4/31X,'E3=',F6.4/31X,
+ 'E4=',F6.4)
C
C  COMPUTE THE 'INEQUALITY PROPORTIONS' U1, U2 AND U3:
      ONE=(SBAR-ABAR)**2
      TWO=(SSTD-ASTD)**2
      THREE=2*(1.-AC(1,1,1))*SSTD*ASTD
      TOT=ONE+TWO+THREE
      U1=ONE/TOT
      U2=TWO/TOT
      U3=THREE/TOT
      WRITE(6,48)
48  FORMAT(/18X,'INEQUALITY PROPORTIONS : ')
      WRITE(6,50) U1,U2,U3
50  FORMAT(/31X,'U1=',F6.4/31X,'U2=',F6.4/31X,'U3=',F6.4)
C
C  COMPUTE THE 'DISCREPANCY PROPORTION' U:
      EBAR=SBAR-ABAR
      ESUMS=0.
      DO 501 I=1,N
      ERRR(I)=S(I)-A(I)
501  ESUMS=ESUMS+(ERRR(I)-EBAR)**2
      ESTD=SQRT(ESUMS/N)
      U=ESTD/(ASTD+SSTD)
      WRITE(6,502) U
502  FORMAT(/18X,'DISCREPANCY COEFFICIENT U =',F6.4)
      IF(NOPT.EQ.0)GO TO 39
C
C  COMPUTE THE STANDARD ERRORS OF THE DIFFERENCES OF THE
C  AUTOCORRELATION FUNCTIONS OF THE OBSERVED AND MODEL-GENERATED
C  BEHAVIOR PATTERNS:
      DO 31 I=1,M
      STDDIF(I)=SQRT(VARARO(I)+VARSRD(I))
      SACV(I)=2.0*STDDIF(I)
31  AACV(I)=SRO(I)-ARO(I)
      CALL CLEAR
C
C  PLOT THE DIFFERENCE CURVE AND THE CONFIDENCE BANDS:
      CALL PLOTTR(SACV,YMAX,M,HOR)
      DO 311 I=1,M
311  SACV(I)=-1*SACV(I)
      CALL PLOTTR(SACV,YMAX,M,HOR)
      CALL PLOTTR(AACV,YMAX,M,STAR)
      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 32 I=2,61
32  WRITE(6,27)(GRAF(I,J),J=1,55)
C
C  THE 2-STANDARD ERROR LIMITS FOR THE DIFFERENCES OF THE PARTIAL
C  AUTOCORRELATION FUNCTIONS:
      TWSIG=2*SQRT(2./N)
      DO 33 I=1,M
      SACV(I)=TWSIG
33  AACV(I)=SPAC(I)-APAC(I)
      CALL CLEAR
C
C  PLOT THE DIFFERENCES OF THE PARTIAL AUTOCORRELATIONS:
      YMAX=0.60
      CALL PLOTTR(SACV,YMAX,M,HOR)
      DO 331 I=1,M
331  SACV(I)=-1*SACV(I)
      CALL PLOTTR(SACV,YMAX,M,HOR)
      CALL PLOTTR(AACV,YMAX,M,STAR)

```

Appendix 2 (continued)

```

      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 34 I=2,61
34    WRITE(6,27)(GRAF(I,J),J=1,55)
      IF(NOPT.EQ.1) GO TO 39
C
C   COMPUTE THE AUTOCORRELATION FUNCTION OF THE RESIDUAL SERIES S-A:
      CALL FTAUTO(ERRR,N,60,50,7,EBAR,EVAR,AACV,ARO,APAC,WAREA)
      WRITE(6,217)
217   FORMAT(1H1//////////20X,'RESIDUAL SERIES = SIMULATED - ',
+ 'OBSERVED')
      WRITE(6,218)
218   FORMAT(//25X,'LAG',5X,'AUCORR',5X,'PAUCORR//')
      DO 215 K=1,M
215   WRITE(6,216) K,ARO(K),APAC(K)
216   FORMAT(25X,I2,2F11,3)
      TWSIG=2*SQRT(1.0/N)
      DO 220 I=1,M
220   SACV(I)=TWSIG
      CALL CLEAR
C
C   PLOT THE AUTOCORRELATIONS OF THE RESIDUALS:
      YMAX=1.0
      CALL PLOTTR(SACV,YMAX,M,HOR)
      DO 221 I=1,M
221   SACV(I)=-1*SACV(I)
      CALL PLOTTR(SACV,YMAX,M,HOR)
      CALL PLOTTR(ARO,YMAX,M,STAR)
      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 230 I=2,61
230   WRITE(6,27)(GRAF(I,J),J=1,55)
      CALL CLEAR
C
C   PLOT THE PARTIAL AUTOCORRELATIONS OF THE RESIDUALS:
      CALL PLOTTR(SACV,YMAX,M,HOR)
      DO 231 I=1,M
231   SACV(I)=-1*SACV(I)
      CALL PLOTTR(SACV,YMAX,M,HOR)
      CALL PLOTTR(APAC,YMAX,M,STAR)
      WRITE(6,259) YMAX,(GRAF(1,J),J=1,5),(L,L=5,M,5)
      DO 240 I=2,61
240   WRITE(6,27)(GRAF(I,J),J=1,55)
39    STOP
      END
C
C   THIS SUBROUTINE COMPUTES THE VARIANCES OF SAMPLE AUTOCORRELATIONS
C
      SUBROUTINE VAUTO(N,N1,RO,VAR)
      DIMENSION RO(351),VAR(50)
      RO(351)=1.0
      DO 20 K=1,50
      SUM=0.
      DO 10 J=1,N1
      KJ=IABS(K-J)
      IF(KJ.EQ.0)KJ=351
      SUM=SUM+(RO(KJ)+RO(K+J)-2*RO(K)*RO(J))**2
10    CONTINUE
20    VAR(K)=SUM/(N*(N+2))
      RETURN
      END
C
C   THIS SUBROUTINE COMPUTES THE FOUR SAMPLE MOMENTS OF
C   THE INPUT DATA SET 'X'.
C
      SUBROUTINE MOMT(N,X,XBAR,XM3,XM4)
      DIMENSION X(351)
      SUM3=0.
      SUM4=0.
      DO 10 I=1,N
      SUM3=SUM3+(X(I)-XBAR)**3
      SUM4=SUM4+(X(I)-XBAR)**4
10    CONTINUE
      ABS3=ABS(SUM3)
      XM3=(ABS3/N)**(1/3.)
      IF(SUM3.LT.0.) XM3=-XM3
      XM4=(SUM4/N)**(1/4.)
      RETURN
      END
C
C   THIS SUBROUTINE FITS SIMPLE LINEAR REGRESSION CURVE TO THE
C   INPUT DATA SET 'X'.
C
      SUBROUTINE REGRES(X,N,BO,B1,VARBO,VARB1)
      DIMENSION X(351)
      SUMX=0.

```

Appendix 2 (continued)

```

SUMNX=0.
DO 100 I=1,N
SUMX=SUMX+X(I)
100 SUMNX=SUMNX+I*X(I)
BO=2*(2*N+1)*SUMX/(N*(N-1)) - 6*SUMNX/(N*(N-1))
B1=12*SUMNX/(N*(N*2-1)) - 6*SUMX/(N*(N-1))
ERSUM=0.
DO 200 I=1,N
XHAT=BO+B1*I
ERSUM=ERSUM+(X(I)-XHAT)**2
200 X(I)=X(I)-B1*I
VARE=ERSUM/(N-2)
VARBO=VARE*2*(2*N+1)/(N*(N-1))
VARB1=VARE*12/(N*(N*2-1))
RETURN
END

C THIS SUBROUTINE PREPARES THE PLOT MATRIX 'GRAF' FOR THE
C INPUT DATA SET 'Y'.
C
SUBROUTINE PLOTTR(Y,AMAX,LAG,SYMB)
DIMENSION Y(351)
COMMON GRAF(61,60)
T=30/AMAX
DO 500 IJ=1,LAG
SGN=ABS(Y(IJ))/(Y(IJ)+0.000001)
ITOP=Y(IJ)*T+0.5*SGN
LE=31-ITOP
500 GRAF(LE,IJ+1)=SYMB
RETURN
END

C THIS SUBROUTINE CLEARS THE PLOT MATRIX 'GRAF', AND
C PREPARES THE HORIZONTAL AND VERTICAL AXES.
C
SUBROUTINE CLEAR
COMMON GRAF(61,60)
DATA HOR,VER,PLUS,CROSS,BLNK/"-","I","+","X"," " /
DO 100 I=1,61
GRAF(I,1)=VER
DO 100 J=2,60
100 GRAF(I,J)=BLNK
DO 200 J=2,60
200 GRAF(31,J)=HOR
DO 300 K=6,46,5
300 GRAF(31,K)=PLUS
DO 400 I=1,61,6
400 GRAF(I,1)=PLUS
RETURN
END

C THIS SUBROUTINE PRINTS THE FREQUENTLY USED TABLE HEADINGS.
C
SUBROUTINE HEAD
WRITE(6,102)
102 FORMAT(1H1//////////////////22X,"AUTOCORRELATION FUNCTIONS",7X,
+"PARTIAL AUTOCORRELATIONS")
WRITE(6,103)
103 FORMAT(/17X,"LAG",4X,"OBSERVED",4X,"SIMULATED",10X,"OBSERVED",
+4X,"SIMULATED"/)
RETURN
END

```

Appendix 3

```

* MARKET DYNAMICS - BY JAY FORRESTER
NOTE
NOTE MINOR MODIFICATIONS BY YAMAN BARLAS, GEORGIA TECH, 1985
NOTE MODIFICATIONS ARE MARKED BY NOTE STATEMENTS
NOTE
OPT CN,SL
OPT PLW=61
FNCTN TEST(5,1)
OPT MSV=251
NOISE 9999999
NOTE
NOTE POSITIVE LOOP - SALESMEN
NOTE
L S.K=S.J+DT*(SH.JK+O) SALESMEN
N S=SN
C SN=10
R OB.KL=S.K*SE.K ORDERS BOOKED

```

Appendix 3 (continued)

```

A B.K=DRA.K*RS    BUDGET
C RS=12    REVENUE TO SALES
A IS.K=B.K/SS    INDICATED SALESMEN
C SS=2000    SALESMEN SALARY
R SH.KL=(1/SAT)*(IS.K-S.K)    SALESMEN HIRED
C SAT=20
NOTE
NOTE NEGATIVE LOOP - MARKET
NOTE
L BL.K=BL.J+DT*(OB.UK-DR.UK)    BACKLOG
N BL=BLN
C BLN=8000
A DDM.K=BL.K/PC.K    DELIVERY DELAY MINIMUM
A PCF.K=TABHL(TPCF,DDM.K,0.5,0.5)    PRODUCTION CAP. FRACT.
T TPCF=0/0.25/0.5/0.67/0.8/0.87/0.93/0.95/0.97/0.98/1.0
NOTE
NOTE THE FOLLOWING NOISE EQUATIONS BY YAMAN BARLAS
R DR.KL=MEANDR.K+SAMPLE(NOIS1.K,INT1,0.0)    DELIVERY RATE
A MEANDR.K=PC.K*PCF.K
C INT1=2.0
A NOIS1.K=NORMRN(0.0,STD1.K)
A STD1.K=S1*MEANDR.K
C S1=0.05
NOTE END OF THIS SET OF MODIFICATIONS
NOTE
L DRA.K=DRA.J+DT*(1/DRAT)*(DR.UK-DRA.J)    DELIVERY RATE AVG.
N DRA=DR
C DRAT=1
A DDI.K=BL.K/DRA.K    DELIVERY DELAY INDICATED
L DDRC.K=DDRC.J+DT*(1/TDDRC)*(DDI.J-DDRC.J)    RECOGN. BY COMPANY
N DDRC=DDI
NOTE
NOTE THE FOLLOWING TDDRC VALUE IS CHANGED FROM 4 TO 2
C TDDRC=2
L DDRM.K=DDRM.J+DT*(1/TDDRM)*(DDRC.J-DDRM.J)    RECOGN. BY MARKET
N DDRC=DDRC
C TDDRM=6
A SEDM.K=TABHL(TSEDM,DDRM.K,0.10,1)    SALES EFFECTIVENESS FROM DELAY
T TSEDM=1/.97/.87/.73/.53/.38/.25/.15/.08/.03/.02
A SE.K=SEDM.K*SEM.K    SALES EFFECTIVENESS
NOTE
NOTE THE FOLLOWING NOISE EQUATIONS BY YAMAN BARLAS
A SEM.K=EMEAN+SAMPLE(NOIS2.K,INT2,0.0)    SALES EFFECTIVENESS MAX.
NOTE
NOTE THE FOLLOWING PARAMETER VALUE IS CHANGED FROM 400 TO 800
C EMEAN=800
C INT2=1.0
A NOIS2.K=NORMRN(0.0,STD2.K)
A STD2.K=S2*EMEAN
C S2=0.05
NOTE END OF THIS SET OF MODIFICATIONS
NOTE
NOTE NEGATIVE LOOP- PRODUCTION CAPACITY
NOTE
L DDT.K=DDT.J+DT*(1/TDDT)*(DDRC.J-DDT.J)    DELIVERY DELAY TRADITIONAL
N DDT=DDRC
C TDDT=12
A DDOG.K=DDT.K    DELIVERY DELAY OPERATING GOAL
A DDC.K=(DDRC.K/DDOG.K)-DDB    DELIVERY DELAY CONDITION
C DDB=.3    DELIVERY DELAY BIAS
A CEF.K=TABHL(TCEF,DDC.K,0.2,5,.5)    CAPACITY EXPANSION FRACTION
NOTE
NOTE THE FOLLOWING TABLE VALUES ARE MODIFIED
T TCEF=-0.03/-0.01/0.0/0.02/0.07/0.15
R PCO.KL=PC.K*CEF.K*ONOFF    PRODUCTION CAPACITY ORDERING
C ONOFF=1    USED TO TURN ON THE PRODUCTION CAP. LOOP
R PCR.KL=DELAY3(PCO.UK,PCRD)    PRODUCTION CAP. RECEIVING
C PCRD=12
L PCDD.K=PCDD.J+DT*(PCO.UK-PCR.UK)    PRODUCTION CAP. ON ORDER
N PCDD=PCO*PCRD
L PC.K=PC.U+DT*(PCR.UK+O)    PRODUCTION CAPACITY
N PC=PCN
C PCN=12000
NOTE
NOTE THE FOLLOWING EQUATIONS ARE ADDED BY YAMAN BARLAS
A OBS.K=OB.UK+SAMPLE(ONOS.K,INT,0.0)    OBSERVED VARIABLE
C ONOS=1.0
A ONOS.K=NORMRN(0.0,OSTD.K)
A OSTD.K=OFAC*OB.UK
C OFAC=0.0
C TO=52
A DUMM.K=TEST(COUNT.K,OBS.K,TIME.K,LENGTH,TO)    DUMMY VAR. TO CALL FORTRAN FUNCT.
NOTE
SPEC DT=0.25/LENGTH=200/PRTPER=10/PLTPER=1/SAVPER=1
SAVE OBS
PLOT OBS=O(0,24000)
PRINT OBS
RUN REAL

```

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